

Copula Modelling for Mixed Continuous and Binary Variables in Survival Analysis: Applications to Colorectal Cancer

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Abstract

Clinical trials investigating cancers increasingly require further analysis into the dependence between tumour progression and survival rates to give informed predictions into survival rates. Although current methods (Kaplan-Meier and augmented-binary) give precise and robust estimators for the survival function, they fail to describe the dependence between variables that affect survival outcomes in patients and miss important structure that could improve future analysis. This paper develops and evaluates a copula-based joint survival function based on continuous and binary variables that arise from clinical trials. A comprehensive simulation study was conducted with the Clayton, Frank and Gumbel copulas to find the best fitting dependence structure, based on coverage rate and average confidence interval width. The results of this study show that the copula-based survival function gives more precise estimates of the survival rate compared to the Kaplan-Meier estimator and increased power in scenarios where tumour and new lesions have an underlying dependence, as well as extra inference through the Kendall's tau and tail dependence. We analysed tumour and survival data from a Phase III colorectal cancer trial and showed the copula method can be applied to real data. These findings highlight the importance of modelling dependence in cancer trials and provide practical guidance for choosing appropriate copula families.

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Chapter 1

Introduction

Clinical trials often track many continuous variables such as blood pressure, tumour size and concentrations of substances in the body which affect a patient. In addition, they also use binary and other discrete variables to describe clinical outcomes or health states, such as whether a patient has diabetes or whether an individual or group is classified as being at risk for a particular disease. These variables may act as indicators of disease status and are collectively referred to as biomarkers. Clinicians focus on when biomarkers pass a given threshold or display a certain indicator. The time at which these thresholds are crossed or indicators occur can define clinically relevant progression events that statisticians model using survival analysis.

Survival analysis is an area of statistics that focuses on time-to-event data and analysing the time taken for a specified event to occur [20]. This is crucial for a wide range of subjects and industries who might want to predict when an event happens, especially in medicine. It can be used to predict when a patient might die or the chances that a clinical trial drug stops the progression of a disease. These examples and many others are important for helping patients and clinicians make informed medical choices.

There are many established methods to predict survival probabilities in these scenarios. Current approaches to modelling survival data include the Kaplan-Meier (KM) estimator, proposed by [17] in 1958 to estimate the survival function and the augmented-binary method [22] which aggregates probability mass or density functions into one survival function. The Cox proportional hazard model and the accelerated failure time model are widely used for estimating the hazard functions of a population and capture different assumptions about the data and its structure [20]. These regression-based methods primarily quantify marginal covariate effects on the hazard or survival function and may not explicitly characterise dependence between progression outcomes.

The copula models are widely used in medical applications [27]. Survival analysis traditionally focuses on a single event of interest which the previously mentioned methods accurately model. However, in a clinical setting, a patient may be subject to several types of event or competing endpoints which can interact with each other, creating dependencies we wish to quantify. Copula-based joint survival functions provide a robust framework for quantifying the underlying dependence between competing events.

Copula theory is an area of statistics that investigates how a collection of functions, called copulas, join multivariate distribution functions to their one dimensional marginal distribution functions [25]. In 1959, Sklar produced Sklar’s Theorem [34] which proved that for any joint distribution function F with univariate marginal distributions, there always exists a copula function C . This result gave us a new way of looking at multivariate distributions. Traditional multivariate distributions have less flexibility compared to copulas which decouple the marginals from the dependence structure. Two different biomarkers can be joined together regardless of their distribution. They are commonly used in medicine, to model variables and biomarkers of patients in order to understand relationships between them or predict survival outcomes.

Existing survival methods often fail to exploit dependence between continuous tumour progression measurements and binary progression indicators, potentially resulting in reduced efficiency and loss of inferential information. This dissertation investigates whether copula-based joint survival models can address these limitations. The proposed method will consider two progression events which are defined under the Response Evaluation Criteria in Solid Tumours (RECIST) [3]. This criteria defines an increase in tumour size of more than 20% compared to the minimum of all previous tumour sizes as a progression event. The appearance of a new lesion is also considered a progression event. The copula model will be used to join these continuous and binary variables together in the context of a colorectal cancer data set which uses RECIST to define when patients progress.

To address this problem, the underlying theory of copula functions and survival analysis must first be introduced. This includes key copula families, measures of dependence and the fundamental concepts of survival modelling. Combining these frameworks allows the construction of a joint survival copula model, which forms the basis of the simulation and real-data analyses.

The simulation study will investigate both single arm and comparative trials. In the simulated single arm trial, the copula method will be compared against the Kaplan-Meier estimator and augmented-binary method based on the average confidence interval width and the coverage rate. In the simulated comparative trial, the power of detecting a difference between the control and treatment arms will be assessed for several different scenarios. The proposed copula method is for improving inference on time-to-event data where the event is defined by progression of continuous and binary events.

The remainder of this dissertation is structured as follows. Chapter 2 introduces important properties and examples of copulas, together with common measures of dependence. Chapter 3 presents key concepts in survival analysis, including longitudinal data, estimators of survival functions and joint survival functions. Chapter 4 defines the proposed models and outlines the methods used to assess treatment effects and statistical power. Chapter 5 contains the simulation studies for both single-arm and comparative trials. Chapter 6 applies the Kaplan–Meier and copula models to a colorectal cancer dataset. Finally, Chapter 7 discusses the main findings, limitations and directions for future research.

Chapter 2

Foundations of Copula Theory

2.1 Introduction

Many statistical problems involve modelling multiple random variables that are not independent. In fields such as medical statistics, understanding and accurately representing dependence structures is crucial for valid inference and prediction. However, traditional multivariate distributions often impose restrictive assumptions on both marginal distributions and dependence structures, limiting their flexibility in practical applications [27].

Copula theory provides a powerful framework for overcoming these limitations by allowing the separation of marginal behaviour from the dependence structure. This makes copulas particularly useful in settings where variables follow different marginal distributions but exhibit non-trivial dependence, such as in survival analysis, where multiple time-to-event outcomes or competing risks may arise.

The aim of this chapter is to develop the theoretical foundation required to model such dependence structures effectively. In particular, we introduce key classes of copulas, including Archimedean and elliptical families, and discuss how measures of dependence can be used to characterise relationships between variables. These tools form the basis for flexible multivariate modelling in later chapters.

The chapter is structured as follows. Section 2.2 defines a copula and outlines its key properties. Section 2.3 introduces the Fréchet–Hoeffding bounds and related extremal copulas. Section 2.4 describes the elliptical family of copulas and their applications, while Section 2.5 focuses on the Archimedean family and common use cases. Section 2.6 presents several measures of dependence used to quantify relationships between variables, and Section 2.7 discusses the choice of copula models and their relative performance in different settings.

2.2 Definitions and Key Properties

Definition 2.2.1. A d -dimensional copula $C : [0, 1]^d \rightarrow [0, 1]$ is a cumulative distribution function (CDF) with uniform marginals which we write $C(\mathbf{u}) = C(u_1, \dots, u_d)$. It satisfies the following properties:

- 1) $C(u_1, \dots, u_d)$ is non-decreasing in each component,
- 2) the i^{th} marginal distribution is obtained by $C(1, \dots, 1, u_i, 1, \dots, 1) = u_i$,
- 3) for $a_i \leq b_i, \mathbb{P}(U_1 \in [a_1, b_1], \dots, U_d \in [a_d, b_d]) \geq 0$. This is the rectangle inequality.

Remark 2.2.2. The rectangle inequality is given by:

$$\sum_{i_1=1}^2 \cdots \sum_{i_d=1}^2 (-1)^{|i|} C(u_{1,i_1}, \dots, u_{d,i_d}) \geq 0,$$

for $u_{j,1} = a_j, u_{j,2} = b_j$.

Example 2.2.3. A bivariate copula satisfies the equality

$$C(a_1, a_2) - C(a_1, b_2) - C(b_1, a_2) + C(b_1, b_2) \geq 0.$$

For a bivariate copula, we have a dimension of $d = 2$. We can rewrite $u_{j,1} = a_j, u_{j,2} = b_j$ as $u_{1,1} = a_1, u_{1,2} = b_1, u_{2,1} = a_2, u_{2,2} = b_2$. So our rectangle inequality becomes

$$\sum_{i_1=1}^2 \sum_{i_2=1}^2 (-1)^{i_1+i_2} C(u_{1,i_1}, u_{2,i_2}).$$

Computing the inner sum and substituting in our values for u gives us

$$\begin{aligned} \sum_{i_1=1}^2 \sum_{i_2=1}^2 (-1)^{i_1+i_2} C(u_{1,i_1}, u_{2,i_2}) &= \sum_{i_1=1}^2 [(-1)^{i_1+1} C(u_{1,i_1}, u_{2,1}) + (-1)^{i_1+2} C(u_{1,i_1}, u_{2,2})] \\ &= \sum_{i_1=1}^2 [(-1)^{i_1+1} C(u_{1,i_1}, a_2) + (-1)^{i_1+2} C(u_{1,i_1}, b_2)] . \end{aligned}$$

Now, we compute the remaining sum and substitute in our values

$$\begin{aligned} \sum_{i_1=1}^2 \sum_{i_2=1}^2 (-1)^{i_1+i_2} C(u_{1,i_1}, u_{2,i_2}) &= (-1)^{1+1} C(u_{1,1}, a_2) + (-1)^{1+2} C(u_{1,1}, b_2) \\ &\quad + (-1)^{2+1} C(u_{1,2}, a_2) + (-1)^{2+2} C(u_{1,2}, b_2) \\ &= C(a_1, a_2) - C(a_1, b_2) - C(b_1, a_2) + C(b_1, b_2) \\ &\geq 0. \end{aligned}$$

The following definition and proposition are required to prove Sklar's Theorem, a fundamental result in copula theory.

Definition 2.2.4. The generalised inverse (quantile function) of the CDF, F , is given by

$$F^{-1}(x) = \inf\{v \in \mathbb{R} : F(v) \geq x\},$$

where $x \in [0, 1]$ is a probability level.

Proposition 2.2.5. If U is uniformly distributed on $[0, 1]$, $U \sim U(0, 1)$, and $F_X(x)$ is a CDF. Then $\mathbb{P}(F^{-1}(U) \leq x) = F_X(x)$.

Proof. Let $X = F^{-1}(U)$. Then,

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F_X(x),$$

since $U \sim U(0, 1)$. □

Theorem 2.2.6. Sklar's Theorem

For a d -dimensional cumulative distribution function F , with marginals $F_1, \dots, F_d$, there exists a copula C such that

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)),$$

for all $x_i \in [-\infty, +\infty]$ and $i = 1, \dots, d$.

Copula functions are defined as a CDF so we also want to consider the probability density function (PDF). This can be obtained the usual way by differentiating the CDF with respect to each random variable. We follow the same process as [15]. By the chain rule, the derivative of the copula becomes

$$\begin{aligned} \frac{\partial^d}{\partial x_1 \cdots \partial x_d} F(x_1, \dots, x_d) &= \frac{\partial^d}{\partial x_1 \cdots \partial x_d} C(F_1(x_1), \dots, F_d(x_d)) \\ f(x_1, \dots, x_d) &= f_1(x_1) \cdots f_d(x_d) c(F_1(x_1), \dots, F_d(x_d)) \\ c(F_1(x_1), \dots, F_d(x_d)) &= \frac{f(x_1, \dots, x_d)}{f_1(x_1) \cdots f_d(x_d)}. \end{aligned}$$

where $c(\cdot)$ is the copula density. Let $u_i = F_i(x_i)$. Then, $x_i = F_i^{-1}(u_i)$. Using this, we can substitute into our formula for the PDF to get a density in terms of u_i .

$$c(u_1, \dots, u_d) = \frac{f(F_1^{-1}(u_1), \dots, F_d^{-1}(u_d))}{f_1(F_1^{-1}(u_1)) \cdots f_d(F_d^{-1}(u_d))}.$$

2.3 The Frechet-Hoeffding Bounds

Definition 2.3.1. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, the co-monotonic copula is defined

$$M(\mathbf{u}) := \min\{u_1, \dots, u_d\}.$$

Definition 2.3.2. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, the independence copula is defined

$$\Pi_d(\mathbf{u}) := u_1 \cdot u_2 \cdots u_d,$$

for independent and identically distributed random vector $\mathbf{u}$ on $U(0, 1)$.

Definition 2.3.3. Let $\mathbf{u} = (u_1, u_2) \in [0, 1]^2$. Then, the counter-monotonic copula is defined

$$W(u_1, u_2) := \max\{u_1 + u_2 - 1, 0\}.$$

Example 2.3.4. For $d \geq 3$, $W(\mathbf{u}) = \max\{\sum_{i=1}^d u_i - d + 1, 0\}$ is not a copula. For $d = 2$, we have the counter-monotonic copula

$$W(u_1, u_2) := \max\{u_1 + u_2 - 1, 0\}.$$

Let us consider part 3 of Definition 2.1.1. We want to show that W does not satisfy the rectangle inequality. Let $d \geq 3$ and R the hyper-rectangle defined $R = [a_i, b_i]^d$ with $a_i, b_i \in [0, 1], \forall i$.

Choose a_i, b_i such that $(d - 2) < da_i < (d - 1) < db_i < d$. Then, the rectangle inequality becomes

$$V = \sum_{i_1=1}^2 \cdots \sum_{i_d=1}^2 (-1)^{d-|\epsilon|} W(c_1^\epsilon, \dots, c_d^\epsilon),$$

where $c_i^\epsilon = \begin{cases} a_i & \text{if } \epsilon = 1 \\ b_i & \text{if } \epsilon = 0 \end{cases}$. If W takes less than $d - 1$ "b"s, then $|\epsilon| < d - 1$ and we have $\sum_{i=1}^d c_i^\epsilon < d - 1$. This means

$$W(\mathbf{u}) = \max\left\{\sum_{i=1}^d c_i^\epsilon - d + 1, 0\right\} = 0.$$

So, if W does not have enough "b"s, then this W has no contribution to the sum. Now, if W has lots of "b"s, $|\epsilon| > d - 1$, then $\sum_{i=1}^d c_i^\epsilon > d - 1$ and $W(\mathbf{u}) = \sum_{i=1}^d c_i^\epsilon - d + 1 > 0$.

All other W that contain 0 or 1, a_i or b_i contribute 0 to the sum. The alternating addition and subtraction, and the fact very few W contribute to the sum, results in over subtraction and some rectangles having $V < 0$, which does not satisfy the rectangle inequality.

Therefore, $W_d(\mathbf{u})$ is not a copula for $d \geq 3$.

Theorem 2.3.5. The Frechet-Hoeffding Bounds

For every copula $C : [0, 1]^d \rightarrow [0, 1]$ and every $\mathbf{u} \in [0, 1]^d$, we have

$$W_d(\mathbf{u}) = \max\left\{\sum_{i=1}^d u_i - d + 1, 0\right\} \leq C(\mathbf{u}) \leq \min\{u_1, \dots, u_d\} = M_d(\mathbf{u}).$$

Proof. Proof of upper bound: C is the CDF of the event $\{U_1 \leq u_1, \dots, U_d \leq u_d\}$. Then $\{U_1 \leq u_1, \dots, U_d \leq u_d\}$ is a subset of each individual event $\{U_i \leq u_i\}$.

So the joint probability cannot be greater than a single marginal. So the smallest upper bound of C is $\min\{u_1, \dots, u_d\}$. We have

$$C(\mathbf{u}) \leq \min\{u_1, \dots, u_d\}.$$

Proof of lower bound: To find the lower bound, we need to consider the complement of $\mathbb{P}(U_1 \leq u_1, \dots, U_d \leq u_d)$.

$$\begin{aligned} \mathbb{P}(U_1 \leq u_1, \dots, U_d \leq u_d) &= \mathbb{P}\left(\bigcap_{i=1}^d \{U_i \leq u_i\}\right) \\ &= 1 - \mathbb{P}\left(\bigcup_{i=1}^d \{U_i > u_i\}\right) \\ &\geq 1 - \sum_{i=1}^d \mathbb{P}(U_i > u_i) \\ &= 1 - \sum_{i=1}^d [1 - \mathbb{P}(U_i \leq u_i)] \\ &= 1 - d + \sum_{i=1}^d u_i. \end{aligned}$$

Probabilities can only be positive or 0, so if the expression above is negative, we assign 0 to the lower bound. Finally, we get the lower bound

$$\max\left\{1 - d + \sum_{i=1}^d u_i, 0\right\} \leq C(\mathbf{u}).$$

The full bounds are now

$$\max\left\{1 - d + \sum_{i=1}^d u_i, 0\right\} \leq C(\mathbf{u}) \leq \min\{u_1, \dots, u_d\}.$$

□

2.4 Elliptic Copulas

One important class of copulas are elliptic copulas. These types are used to capture linear dependence, not just correlation. They are symmetric and work well when variables display roughly equal co-movements. The two main types that we will look at are Gaussian and t-copulas.

2.4.1 Gaussian Copula

Definition 2.4.1. Gaussian Copula

Let $\mathbf{X} = \{X_1, \dots, X_d\} \in \mathbb{R}^d$ be a random vector with distribution $\mathbf{X} \sim N_d(\mathbf{0}, \Sigma)$ where Σ is the linear correlation matrix. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, $C_\Sigma^G(\mathbf{u})$ is the distribution function of $(\Phi(X_1), \dots, \Phi(X_d))$ where Φ is the univariate standard normal distribution. We write

$$\begin{aligned} C_\Sigma^G(\mathbf{u}) &= \mathbb{P}(\Phi(X_1) \leq u_1, \dots, \Phi(X_d) \leq u_d), \\ &= \Phi_\Sigma^d(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)). \end{aligned}$$

Figure 2.1 shows the distribution of the Gaussian copula with both strong/weak correlation and positive/negative correlation. As the correlation parameter ρ increases, the mass of the copula becomes increasingly concentrated around the diagonal $u_1 = u_2$, reflecting stronger positive dependence between the variables. For negative values of ρ , the dependence structure shifts towards the other diagonal, indicating that larger values of one variable tend to correspond to smaller values of the other.

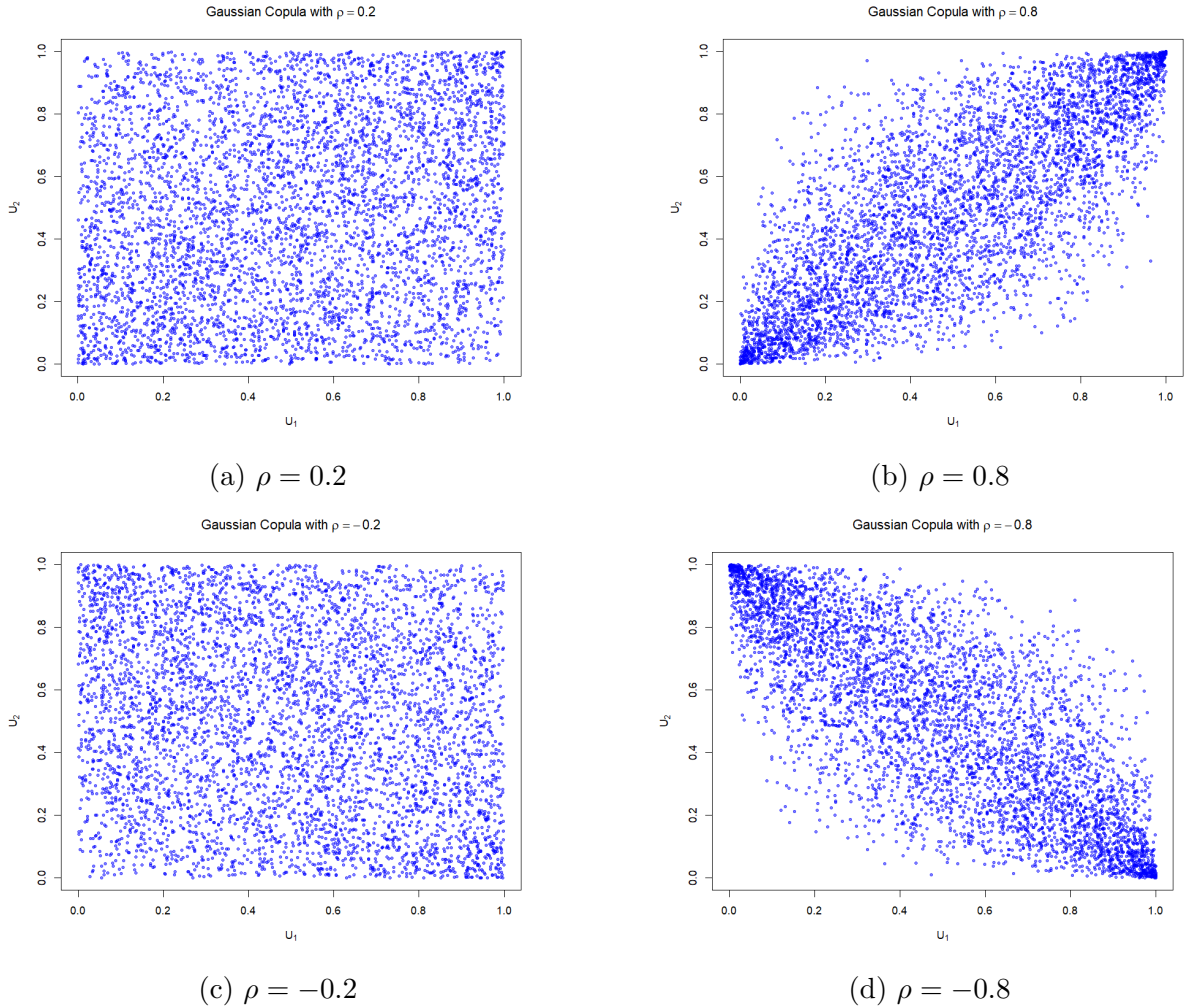


Figure 2.1: Gaussian copulas with different parameters ρ .

Example 2.4.2. We wish to find the joint density function of $Y_1 \sim \text{Exp}(\lambda_1)$ and $Y_2 \sim N(\mu_2, \sigma_2^2)$ using the Gaussian copula.

In Section 2.2, we derived the formula for the density of a copula. In this bivariate case,

it is easier to use the untransformed version

$$f_{Y_1, Y_2}(y_1, y_2) = f_1(y_1)f_2(y_2)c(F_1(x_1), F_2(x_2)).$$

We know the density of Y_1 and Y_2 .

$$f_{Y_1}(y_1) = \begin{cases} \lambda \exp^{-\lambda y_1}, & \text{if } y_1 \geq 0, \\ 0, & \text{if } y_1 < 0. \end{cases}$$

$$f_{Y_2}(y_2) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(y_2 - \mu_2)^2}{2\sigma_2^2}\right), & \text{if } y_2 \in \mathbb{R}, \\ 0, & \text{otherwise.} \end{cases}$$

We need the cumulative distribution functions of Y_1 and Y_2 . We obtain these by integrating the density to get

$$F_{Y_1}(y_1) = 1 - \exp^{-\lambda y_1}, \quad F_{Y_2}(y_2) = \Phi(y_2; \mu_2, \sigma_2^2).$$

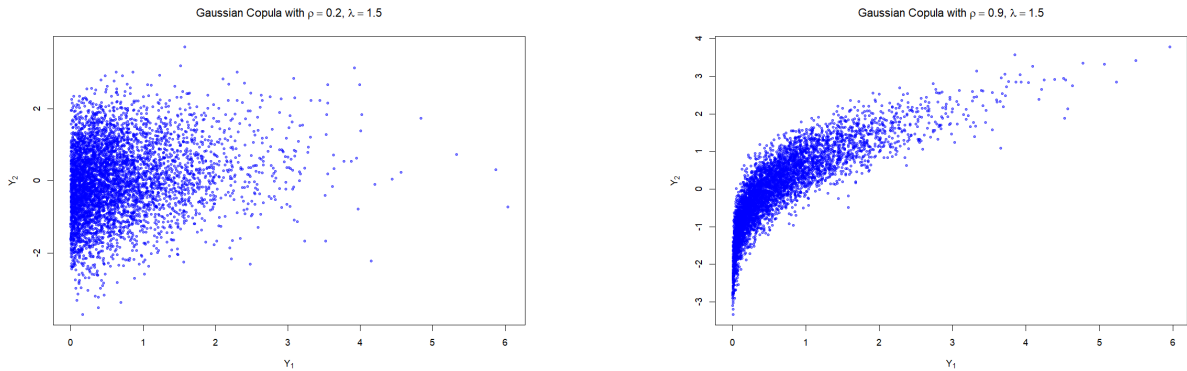
Finally, we need to standardise Y_1 and Y_2 . We do this with the transformation $z_k = \frac{y_k - \mu_k}{\sigma_k}$ for $k = 1, 2$. Let $u_1 = \Phi(z_1)$, $u_2 = \Phi(z_2)$. Then, the density of the Gaussian copula is given by

$$c(z_1, z_2; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}(z_1^2 - 2\rho z_1 z_2 + z_2^2)\right).$$

Combining parts gives us the final joint density distribution of Y_1 and Y_2 .

$$f_{Y_1, Y_2}(y_1, y_2) = (\lambda \exp^{-\lambda y_1}) \left(\frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(y_2 - \mu_2)^2}{2\sigma_2^2}\right) \right) c(1 - \exp^{-\lambda y_1}, \Phi(y_2; \mu_2, \sigma_2^2); \rho).$$

Figure 2.2 shows this joint distribution with a weak correlation and a strong correlation. With $\rho = 0.2$, the joint distribution is diffuse and only shows a very weak increasing curve, whereas with $\rho = 0.9$ the points concentrate along a clear increasing curve, indicating strong dependence. In both cases the dependence remains symmetric and exhibits no tail behaviour, reflecting the elliptical nature of the Gaussian copula.



(a) $\rho = 0.2$, $\lambda = 1.5$

(b) $\rho = 0.9$, $\lambda = 1.5$

Figure 2.2: Gaussian copula with exponential and normal marginals.

Remark 2.4.3. Sometimes, we may want to combine discrete and continuous marginals in a copula. It is tricky to do this without some work on the discrete variable. If we introduce a latent variable, we can model the discrete variable as a continuous variable, making it much easier to combine the two marginals and get a well-behaved copula.

Example 2.4.4. Consider the joint density of $Y_1 \sim \text{Bin}(n, p)$ and $Y_2 \sim \text{Exp}(\lambda)$ using a Gaussian copula.

Again, we will use the following formula to get the joint density of Y_1 and Y_2 .

$$f_{Y_1, Y_2}(y_1, y_2) = f_1(y_1)f_2(y_2)c(F_1(x_1), F_2(x_2)).$$

Y_2 is a binomial random variable which takes values in $k = 0, \dots, n$. We can define a standard normal random variable Z_2 , which takes value k in a pre-specified interval.

$$Y_2 = \begin{cases} 0, & a_{-1} \leq Z_2 \leq a_0, \\ 1, & a_0 \leq Z_2 \leq a_1, \\ \cdot & \\ \cdot & \\ \cdot & \\ n-1, & a_{n-2} \leq Z_2 \leq a_{n-1}, \\ n, & a_{n-1} \leq Z_2 \leq a_n. \end{cases}$$

where $a_{-1} = -\infty$, $a_n = +\infty$ and $a_0, \dots, a_{n-1} \in \mathbb{R}$.

From this we can derive the CDF of Y_2 in terms of the latent variable Z_2 . A basic fact from probability about discrete CDFs is that

$$\mathbb{P}(Y_2 \leq k) = \mathbb{P}(Y_2 = 0) + \mathbb{P}(Y_2 = 1) + \dots + \mathbb{P}(Y_2 = k-1) + \mathbb{P}(Y_2 = k).$$

We have defined Y_2 in a way such that it is dependent on Z_2 . So, we can rewrite the equation above in terms of the standard normal CDF and a_k .

$$\begin{aligned} \mathbb{P}(Y_2 \leq k) &= \mathbb{P}(Y_2 = 0) + \mathbb{P}(Y_2 = 1) + \dots + \mathbb{P}(Y_2 = k-1) + \mathbb{P}(Y_2 = k), \\ &= \mathbb{P}(a_{-1} \leq Z_2 \leq a_0) + \mathbb{P}(a_0 \leq Z_2 \leq a_1) + \dots + \mathbb{P}(a_{k-1} \leq Z_2 \leq a_k), \\ &= [\Phi(a_0) - \Phi(a_{-1})] + [\Phi(a_1) - \Phi(a_0)] + \dots + [\Phi(a_k) - \Phi(a_{k-1})], \\ &= \Phi(a_k) - \Phi(a_{-1}). \end{aligned}$$

So, $F_2(k) = \mathbb{P}(Y_2 \leq k) = \Phi(a_k)$ since we defined $a_{-1} = -\infty$ and $\Phi(-\infty) = 0$. We can easily derive $f_2(y_2)$ by differentiating $F_2 = \Phi(a_k)$. We get $f_2(y_2) = \phi(a_k)$. Now, let

$$\begin{aligned} u_1 &= \Phi(z_1), & u_2 &= \Phi(z_2), \\ u_1 &= F_1(y_1), & u_2 &= F_2(y_2). \end{aligned}$$

Then,

$$z_1 = \Phi^{-1}(F_1(y_1)), \quad z_2 = z_2.$$

From Example 2.4.2, we already have the density of the Gaussian copula, f_{Y_1} , and F_{Y_1} so we can combine each part to get the joint probability density

$$f_{Y_1, Y_2}(y_1, z_2) = c(1 - \exp(-\lambda y_1), z_2) \cdot \lambda e^{-\lambda y_1} \cdot \phi(z_2).$$

2.4.2 t-Copula

Definition 2.4.5. t-Copula

Let $\mathbf{X} = \{X_1, \dots, X_d\} \in \mathbb{R}^d$ be a random vector with distribution $\mathbf{X} \sim \mathbf{t}_d(\nu, \mathbf{0}, \mathbf{P})$ where ν is the degrees of freedom and $\mathbf{P}$ is the correlation matrix. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then $C_{\nu, \rho}^t(\mathbf{u})$ is the distribution function of $(t_\nu(X_1), \dots, t_\nu(X_d))$. We write

$$\begin{aligned} C_{\nu, \rho}^t(\mathbf{u}) &= \mathbb{P}(t_\nu(X_1) \leq u_1, \dots, t_\nu(X_d) \leq u_d), \\ &= t_\nu^d(t_\nu^{-1}(u_1), \dots, t_\nu^{-1}(u_d)). \end{aligned}$$

Figure 2.3 shows the t-copula distribution for different correlation parameters and degrees of freedom. Looking down a column where ρ is held constant, increasing the degrees of freedom reduces the heaviness of the tails and the extreme co-movements become less pronounced; the joint distribution becomes more elliptical and approaches the Gaussian copula structure. Looking across a row with fixed ν and increasing ρ , the points align more strongly along the diagonal, indicating stronger overall dependence. The persistence of extreme points in the corners for small ν highlights the lower and upper tail dependence characteristic of the t-copula.

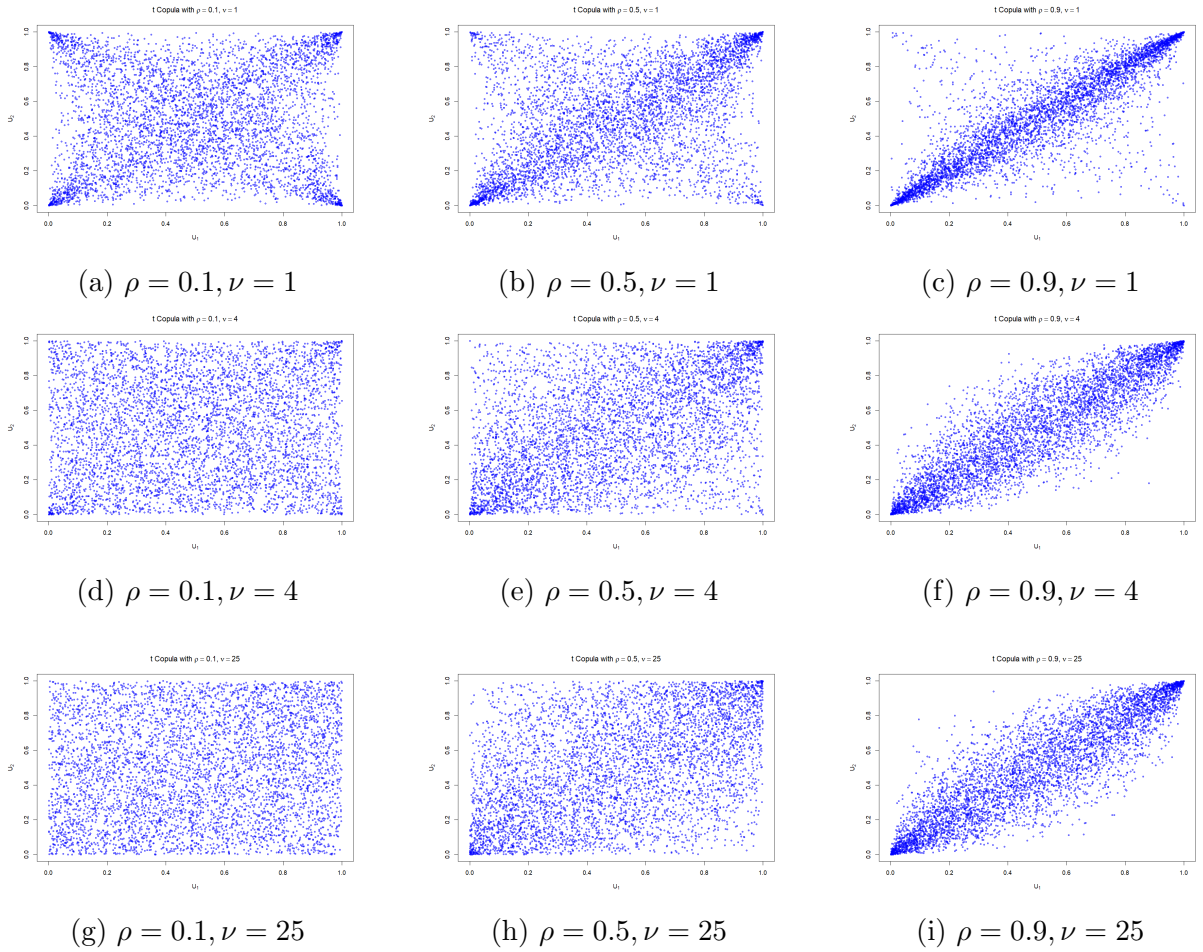


Figure 2.3: t-Copulas with different parameters ρ, ν .

Example 2.4.6. Consider Example 2.4.2 again. Instead of a Gaussian distribution for the copula, we will use a t distribution to find the joint density of exponential and normal variables. The formula for the density is

$$f_{Y_1, Y_2}(y_1, y_2) = f_1(y_1)f_2(y_2)c(F_1(x_1), F_2(x_2)).$$

Similarly, $z_1 = t_\nu^{-1}(1 - \exp(-\lambda y_1))$, $z_2 = t_\nu^{-1}(\Phi(y_2; \mu_2, \sigma_2^2))$. Then, the joint density of these two variables is

$$f_{Y_1, Y_2}(y_1, y_2) = (\lambda \exp^{-\lambda y_1}) \left(\frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(y_2 - \mu_2)^2}{2\sigma_2^2}\right) \right) \\ \times c(t_\nu^{-1}(1 - \exp(-\lambda y_1)), t_\nu^{-1}(\Phi(y_2; \mu_2, \sigma_2^2))),$$

where

$$c_{2,\rho,\nu}(z_1, z_2) = \frac{\Gamma(\frac{\nu+2}{2})}{\Gamma(\frac{\nu}{2}) \nu \pi \sqrt{1-\rho^2}} \left[1 + \frac{1}{\nu(1-\rho^2)} (z_1^2 - 2\rho z_1 z_2 + z_2^2) \right]^{-\frac{\nu+2}{2}},$$

is the bivariate density of the t copula. Figure 2.4 shows the joint distribution with a fixed degrees of freedom $\nu = 5$ and two different correlation parameters.

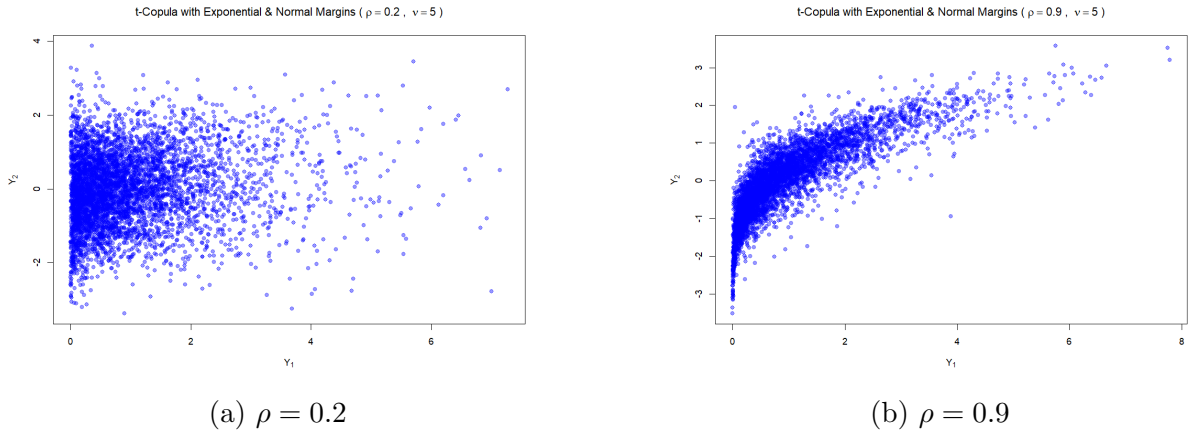


Figure 2.4: t copula with exponential and normal marginals.

2.5 Archimedean Copulas

The Archimedean family of copulas are a simple and flexible way to model non-linear dependence. They are formed by a generator ϕ , where ϕ is a convex, decreasing function such that $\phi : (0, 1] \rightarrow [0, \infty)$ and $\phi(1) = 0$. The general form of the copula is

$$C(u_1, \dots, u_d) = \phi^{-1}(\phi(u_1) + \dots + \phi(u_d)),$$

where $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$.

2.5.1 Clayton Copula

Definition 2.5.1. Clayton Copula

Let $\theta \in [-1, \infty) \setminus \{0\}$. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, the Clayton copula has the form

$$C_d^C(\mathbf{u}; \theta) = \max \left\{ \left(\sum_{i=1}^d u_i^{-\theta} - d + 1 \right)^{-\frac{1}{\theta}}, 0 \right\},$$

where the generator is $\phi(t) = t^{-\theta} - 1$.

However, if we restrict to $\theta > 0$, the Clayton copula simplifies to

$$C^d(\mathbf{u}; \theta) = \left(\sum_{i=1}^d u_i^{-\theta} - d + 1 \right)^{-\frac{1}{\theta}}.$$

Figure 2.5 shows the Clayton copula for two different copula parameter values. As θ increases from 2 to 10, the dependence becomes stronger and the points cluster more tightly in the bottom-left corner. This reflects the lower tail dependence of the Clayton family. Small values of U_1 and U_2 tend to occur together. In contrast, the upper-right region remains relatively diffuse, highlighting the absence of upper tail dependence.

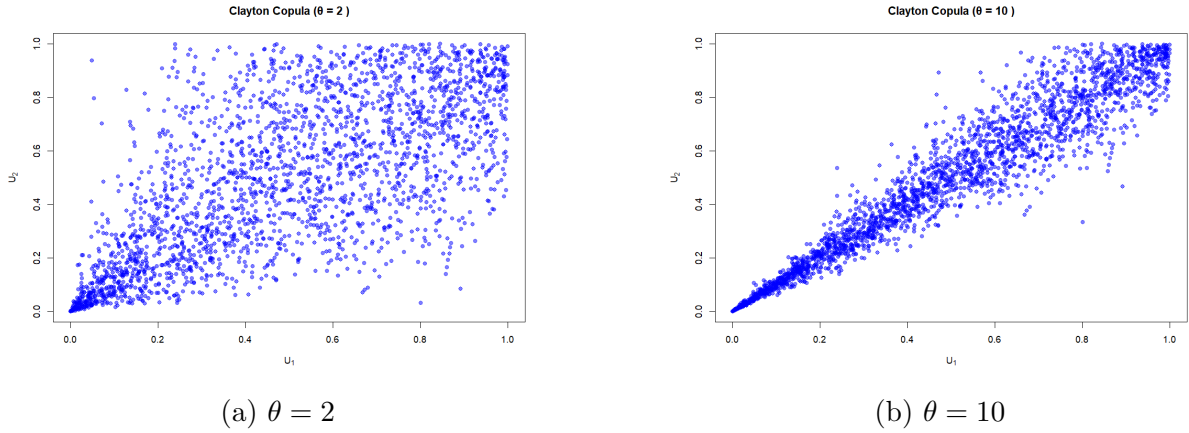


Figure 2.5: Clayton copulas with different parameters θ .

Example 2.5.2. Let us find the joint density of $Y_1 \sim \text{Exp}(\lambda_1)$ and $Y_2 \sim \text{Gamma}(r, \lambda_2)$ with a Clayton copula. The density functions are given by

$$f_{Y_1}(y_1) = \begin{cases} \lambda_1 \exp^{-\lambda_1 y_1}, & \text{if } y_1 \geq 0, \\ 0, & \text{if } y_1 < 0. \end{cases}$$

$$f_{Y_2}(y_2) = \begin{cases} \frac{\lambda_2^r}{(r-1)!} y_2^{r-1} e^{-\lambda_2 y_2}, & y > 0, \\ 0, & y \leq 0. \end{cases}$$

We are working in the bivariate case so we can simplify the Clayton copula to

$$C_2^C(u_1, u_2; \theta) = (u_1^{-\theta} + u_2^{-\theta} - 1)^{-\frac{1}{\theta}}.$$

Now, taking the partial derivative with respect to u_1 and u_2 will give us the density of the Clayton copula.

$$c(u_1, u_2; \theta) = \frac{\partial^2 C_2^C}{\partial u_1 \partial u_2} = (u_1 u_2)^{-(\theta+1)} (u_1^{-\theta} + u_2^{-\theta} - 1)^{-\frac{1+2\theta}{\theta}}.$$

Now, let $u_1 = F_{Y_1}(y_1)$ and $u_2 = F_{Y_2}(y_2)$. Using the following formula, we can get the joint density of Y_1 and Y_2

$$f_{Y_1, Y_2}(y_1, y_2) = f_1(y_1) f_2(y_2) c(F_1(y_1), F_2(y_2)),$$

$$= [\lambda_1 \exp^{-\lambda_1 y_1}] \left[\frac{\lambda_2^r}{(r-1)!} y_2^{r-1} e^{-\lambda_2 y_2} \right] c \left(1 - \exp^{-\lambda_1 y_1}, \frac{\gamma(r, \lambda_2 y_2)}{\Gamma(r)}; \theta \right).$$

where $\gamma(r, \lambda y_2)$ is the lower incomplete gamma function defined

$$\gamma(r, \lambda y_2) = \int_0^{\lambda y_2} t^{r-1} e^{-t} dt.$$

Figure 2.6 shows the Clayton copula for marginals $Y_1 \sim \text{Exp}(1)$ and $Y_2 \sim \text{Gamma}(2, 2)$ with different copula parameters.

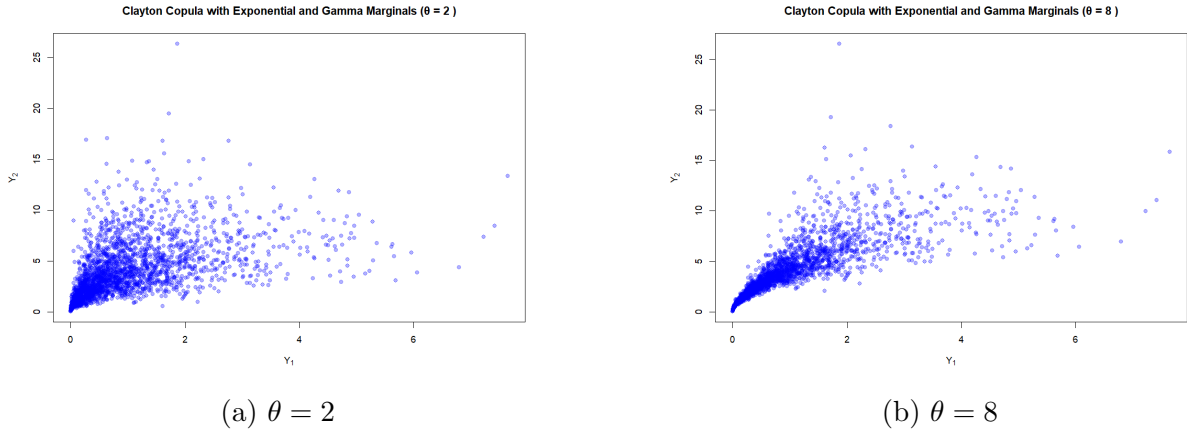


Figure 2.6: Clayton copulas with exponential and gamma marginals.

2.5.2 Gumbel Copula

Definition 2.5.3. Gumbel Copula

Let $\theta \geq 0$. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, the Gumbel copula has the form

$$C_d^{\text{Gum}}(\mathbf{u}; \theta) = \exp \left(- \left(\sum_{i=1}^d (-\log[u_i])^\theta \right)^{\frac{1}{\theta}} \right),$$

with generator $\phi(t) = (-\log(t))^\theta$.

Figure 2.7 shows the Gumbel copula for two different copula parameters. As θ increases from 2 to 10, the dependence strengthens and the points cluster more tightly in the upper right. This reflects the upper-tail dependence of the Gumbel family. Large values of U_1 and U_2 tend to occur together as θ grows. Conversely, the lower-left region remains relatively sparse highlighting the absence of lower tail dependence.

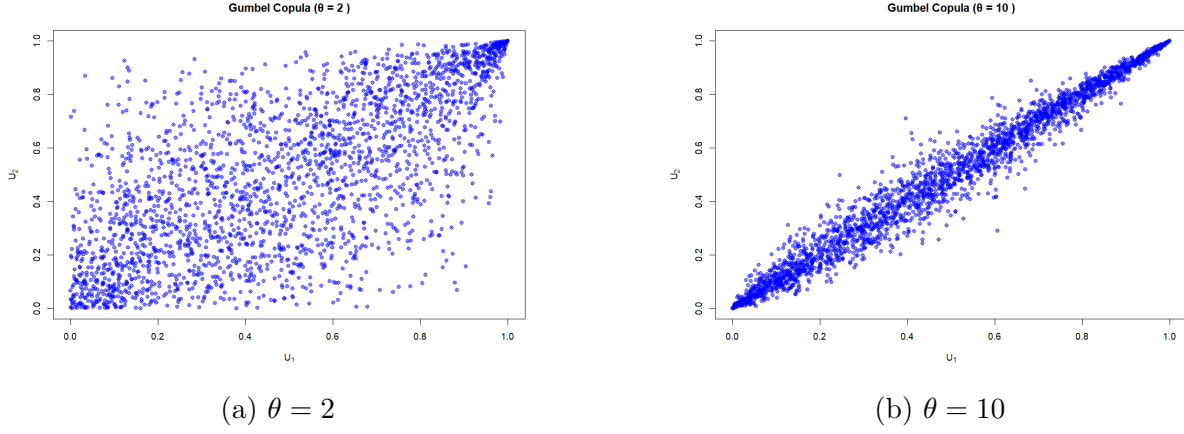


Figure 2.7: Gumbel copulas with different parameters θ .

2.5.3 Frank Copula

Definition 2.5.4. Frank Copula

Let $\theta \in \mathbb{R}$. Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$. Then, the Frank copula has the form

$$C_d^{Fr}(\mathbf{u}; \theta) = -\frac{1}{\theta} \log \left(1 + \frac{\prod_{i=1}^d (e^{-\theta u_i} - 1)}{e^{-\theta} - 1} \right),$$

with generator $\phi(t) = -\log \left(\frac{e^{-\theta t} - 1}{e^{-\theta} - 1} \right)$.

Figure 2.8 shows the Frank copula with two different copula parameters. As θ increases from 2 to 15, the dependence strengthens and the points become more concentrated along the diagonal, indicating stronger overall association between U_1 and U_2 . Unlike the Clayton and Gumbel copulas, the Frank copula is symmetric and does not exhibit tail dependence so extreme values in either tail occur independently.

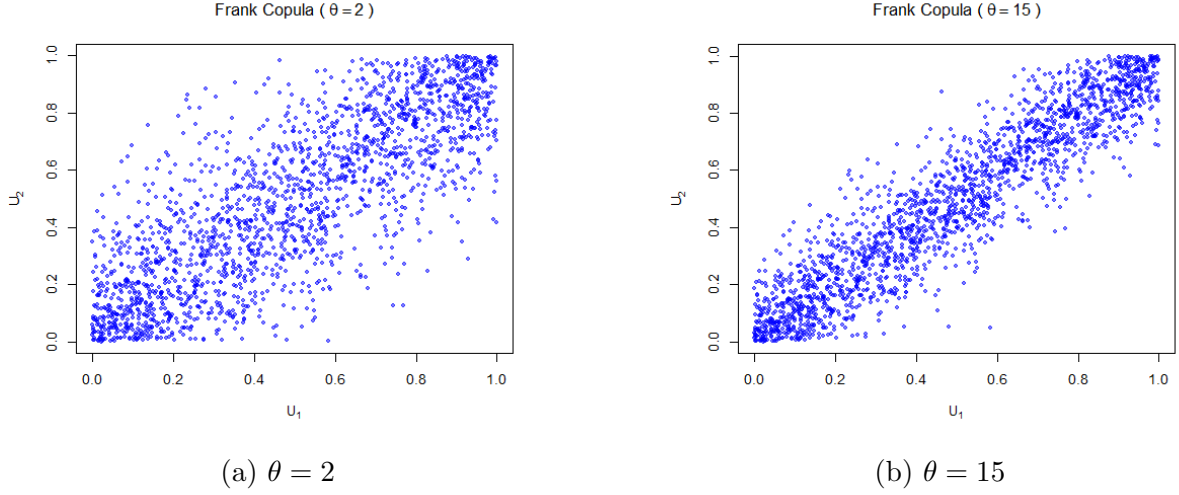


Figure 2.8: Frank copulas with different parameters θ .

2.6 Measures of Dependence

We want to measure the relationship between variables to see how much they relate to each other or to find variables that are correlated or uncorrelated. We will look at three different measures; Pearson's product moment correlation coefficient, Kendall's tau and tail dependence. All these measures tell us useful things about the copula and the data it models.

2.6.1 Pearson's Product Moment Correlation Coefficient

The most widely used measure of correlation is Pearson's product moment correlation coefficient (PMCC). It only measures linear dependence between two variables and is symmetric [38].

Definition 2.6.1. Let X_1 and X_2 be random variables. Then, Pearson's product moment correlation coefficient is defined

$$\rho_{X_1, X_2} = Corr(X_1, X_2) = \frac{Cov(X_1, X_2)}{\sqrt{Var(X_1)Var(X_2)}},$$

where $Cov(X_1, X_2) = \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2]$.

We always have $-1 \leq \rho_{X_1, X_2} \leq 1$ where -1 measures perfect negative correlation and 1 measures perfect positive correlation. It is a simple and intuitive measure to understand but may not be best suited to describing copulas. It is sensitive to outliers and cannot capture non-linear structure in the data which is often the case in complex and high dimensional data sets.

2.6.2 Kendall's Tau

Kendall's τ measures concordance which is a measure of dependence [38]. This tells us if "large" values are associated with other "large" values and if "small" values are associated with "small" values. Much like Pearson's coefficient, ρ_τ is symmetric and we always have $-1 \leq \rho_\tau \leq 1$.

Definition 2.6.2. Let (X_1, Y_1) and (X_2, Y_2) be independent and identically distributed random vectors. Then Kendall's τ is defined

$$\rho_\tau = \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) - \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) < 0).$$

Remark 2.6.3. If X_1 and X_2 both have continuous margins, then we can write Kendall's tau as

$$\rho_\tau = 4 \int \int_{[0,1]^2} C(u, v) dC(u, v) - 1.$$

This result is stated in [38] and we can derive it as follows. First, we expand the expression for ρ_τ .

$$\begin{aligned} \rho_\tau &= \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) - \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) < 0) \\ &= \mathbb{P}(X_1 > X_2, Y_1 > Y_2) + \mathbb{P}(X_1 < X_2, Y_1 < Y_2) \\ &\quad - \mathbb{P}(X_1 > X_2, Y_1 < Y_2) - \mathbb{P}(X_1 < X_2, Y_1 > Y_2). \end{aligned}$$

Since (X_1, Y_1) and (X_2, Y_2) are *iid*, we can note that

$$\begin{aligned} \mathbb{P}(X_1 > X_2, Y_1 > Y_2) &= \mathbb{P}(X_1 < X_2, Y_1 < Y_2) \\ \mathbb{P}(X_1 > X_2, Y_1 < Y_2) &= \mathbb{P}(X_1 < X_2, Y_1 > Y_2). \end{aligned}$$

From this, we can simplify this to

$$\rho_\tau = 2\mathbb{P}(X_1 > X_2, Y_1 > Y_2) - 2\mathbb{P}(X_1 > X_2, Y_1 < Y_2).$$

Also, note that $\mathbb{P}(X_1 > X_2, Y_1 > Y_2) + \mathbb{P}(X_1 > X_2, Y_1 < Y_2) = 1$. So, we can reduce the expression for ρ_τ to just terms of $\mathbb{P}(X_1 > X_2, Y_1 > Y_2)$. We get

$$\begin{aligned} \rho_\tau &= 2\mathbb{P}(X_1 > X_2, Y_1 > Y_2) - 2(1 - \mathbb{P}(X_1 > X_2, Y_1 < Y_2)) \\ &= 4\mathbb{P}(X_1 > X_2, Y_1 > Y_2) - 1. \end{aligned}$$

Now, we need to transform our variables into copula form. Let $U_i = F_X(X_i)$ and $V_i = F_Y(Y_i)$. Then, we get

$$\begin{aligned} \rho_\tau &= 4\mathbb{P}(F_X^{-1}(U_1) > F_X^{-1}(U_2), F_Y^{-1}(V_1) > F_Y^{-1}(V_2)) - 1 \\ &= 4\mathbb{P}(U_1 > U_2, V_1 > V_2) - 1. \end{aligned}$$

We now condition on (U_1, V_1) .

$$\begin{aligned} \rho_\tau &= 4\mathbb{P}(U_1 > U_2, V_1 > V_2) - 1 \\ &= 4\mathbb{E}[\mathbb{P}(U_2 < U_1, V_2 < V_1 | U_1, V_1)] - 1 \\ &= 4\mathbb{E}[C(U_1, V_1)] - 1. \end{aligned}$$

We assumed that we have continuous margins and (U_1, V_1) is distributed by $C(u, v)$. Then, the expectation becomes a double integral and we get

$$\rho_\tau = 4 \int \int_{[0,1]^2} C(u, v) dC(u, v) - 1.$$

2.6.3 Tail Dependence

When comparing two random variables X_1 and X_2 , it is useful to find out how variables behave in the extremes of the distribution. So, given X_1 is extreme, how likely is it that X_2 is extreme? Tail dependence gives us a way to describe this. [38]

Definition 2.6.4. Let X_1 and X_2 be random variables with CDFs F_1 and F_2 respectively.

The coefficient of upper tail dependence, λ_u , is defined

$$\lambda_u := \lim_{q \rightarrow 1} \mathbb{P}(X_2 > F_2^{-1}(q) | X_1 > F_1^{-1}(q)).$$

The coefficient of lower tail dependence, λ_l , is defined

$$\lambda_l := \lim_{q \rightarrow 0} \mathbb{P}(X_2 \leq F_2^{-1}(q) | X_1 \leq F_1^{-1}(q)).$$

Remark 2.6.5. If $\lambda_u > 0$, then X_1 and X_2 have upper tail dependence. Also, if $\lambda_u = 0$, X_1 and X_2 are asymptotically independent in the upper tail. The conclusions are similar for lower tail dependence. This is crucial for determining if a system or model will fail at the same time.

Example 2.6.6. In [26] Exercise 5.47, we are asked to verify the tail dependencies of some different types of copulas in Example 5.22. We now choose an interesting selection to verify.

1) The Archimedean copula C with generator ϕ given by:

$$C(u, v) = \frac{uv}{1 - \theta(1-u)(1-v)} \text{ and } \phi(t) = \log \left(\frac{1 - \theta(1-t)}{t} \right),$$

has $\lambda_U = \lambda_L = 0$.

Solution: First, we use the formula given to us in Theorem 5.42 of [26]

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \left(\frac{1 - C(t, t)}{1 - t} \right).$$

Let us start by substituting t into C .

$$C(t, t) = \frac{t^2}{1 - \theta(1-t)^2}.$$

Then,

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \left(\frac{1 - C(t, t)}{1 - t} \right) = 2 - \lim_{t \rightarrow 1^-} \left(\frac{1 - \theta(1-t)^2 - t^2}{1 - \theta(1-t)^2} \cdot (1-t) \right).$$

Expanding out the limit gives

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \frac{1 - \theta + 3\theta t - 2(\theta + 1)t^2 + 3t^3}{1 - \theta(1-t)^2}.$$

Taking the limit $t \rightarrow 1^-$

$$\lambda_U = 2 - 2 = 0,$$

as required. The lower tail dependence has the formula

$$\lambda_L = \lim_{t \rightarrow 0^+} \left(\frac{C(t, t)}{t} \right).$$

We have already worked out $C(t, t)$ so we can substitute this straight into this formula.

$$\lambda_L = \lim_{t \rightarrow 0^+} \left(\frac{C(t, t)}{t} \right) = \lim_{t \rightarrow 0^+} \left(\frac{t^3}{1 - \theta(1 - t)^2} \right).$$

We can simply take the limit $t \rightarrow 0^+$ and our result is $\lambda_L = 0$ as required.

2) The Archimedean copula C with generator ϕ given by:

$$C(u, v) = \left(1 + ((u^{-1} - 1)^\theta + (v^{-1} - 1)^\theta)^{\frac{1}{\theta}} \right)^{-1} \text{ and } \phi(t) = (t^{-1} - 1)^\theta,$$

has $\lambda_U = 2 - 2^{\frac{1}{\theta}}$, $\lambda_L = 2^{-\frac{1}{\theta}}$.

Solution: Again, we use the formulas provided to find the tail dependence. We start with the lower tail dependence. Firstly,

$$C(t, t) = \left(1 + (2(t^{-1} - 1)^\theta)^{\frac{1}{\theta}} \right)^{-1}.$$

Then,

$$\begin{aligned} \lambda_L &= \lim_{t \rightarrow 0^+} \left(\frac{C(t, t)}{t} \right) \\ &= \lim_{t \rightarrow 0^+} \frac{1}{t(1 + 2^{\frac{1}{\theta}}(t^{-1} - 1))} \\ &= \lim_{t \rightarrow 0^+} \frac{1}{t + 2^{\frac{1}{\theta}}(1 - t)}. \end{aligned}$$

Taking the limit $t \rightarrow 0^+$, we get $\lambda_L = 2^{-\frac{1}{\theta}}$. Now, we consider the upper tail dependence. We know $C(t, t)$, so we can simply substitute this into our formula for the upper tail dependence.

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \left(\frac{1 - C(t, t)}{1 - t} \right) = 2 - \lim_{t \rightarrow 1^-} \frac{1 - \left(1 + (2(t^{-1} - 1)^\theta)^{\frac{1}{\theta}} \right)^{-1}}{1 - t}.$$

By rearranging, we obtain

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \frac{2^{\frac{1}{\theta}}(t^{-1} - 1)}{1 + 2^{\frac{1}{\theta}}(t^{-1} - 1)} \cdot \frac{1}{1 - t} = 2 - \lim_{t \rightarrow 1^-} \frac{2^{\frac{1}{\theta}}}{1 + 2^{\frac{1}{\theta}}(t^{-1} - 1)}.$$

Taking the limit $t \rightarrow 1^-$, we get $\lambda_U = 2 - 2^{\frac{1}{\theta}}$.

2.7 Comparing Copulas

Different problems in reality generate all sorts of datasets that have varying underlying structures so there is no single copula that will be perfect for all scenarios. Here, we will discuss the similarities and differences between families of copulas, the differences in the families themselves and where they might be appropriate.

2.7.1 Comparing Families of Copulas

When considering what copula to use for a problem, we should consider if we are going to use an Archimedean or elliptic copula. Elliptic copulas are made up of multiple elliptic distributions such as the Student's t-distribution or the Gaussian distribution and are defined by a correlation matrix. They are perfect for high dimensional problems and can easily show us the correlation between each variable in the correlation matrix. Archimedean copulas are formed by a generator function, so give a more rigid form that depends on a single parameter. This does not work as well in a high dimensional setting.

However, elliptic copulas can have less flexibility in some sense. With Gaussian and t-copulas, we either have upper and lower tail dependence, or no tail dependence at all. This means if we know how our problem is structured (if there is upper, lower or no tail dependence), then an Archimedean copula could fit that type of data better.

These are the issues we have to weigh up when selecting what model to use.

2.7.2 Comparing Elliptic Copulas

The t copula and Gaussian copula are both elliptical and have a correlation parameters but the way they capture data is very different. The t-copula has two parameters ν , the degrees of freedom and ρ , the correlation. It is a great way to capture when extreme events occur together. For example, say a patient has an extremely large tumour size which causes an extremely low survival probability, the t-copula can identify these low and high extremes together.

The degrees of freedom controls how "fat" the tails of the copula are. Low values of ν give rise to "fat" tails and strong tail dependence whereas large values of ν give weak tail dependence.

This extra parameter gives us more flexibility when we try to fit our copula to a dataset. However, if we want to estimate ν and ρ , this can be computationally expensive and take a long time.

In contrast, the Gaussian copula is one of the most widely computed distributions available with only the correlation parameter ρ to estimate. It has zero tail dependence so events in the tails of the distribution behave like they are independent.

The Gaussian copula is fast, stable and widely used making it a great start when first

looking at a problem but due to it's lack of tail dependence, it can miss extreme events that coincide with each other. The t-copula has a greater flexibility with ν and ρ to capture the structure and dependency of a problem, even if it comes at a computational cost. We also have the interesting property that if we let $\nu \rightarrow \infty$, the t-copula converges to the Gaussian copula. We will show this relationship below.

Example 2.7.1. We want to show the t-copula converges to the Gaussian copula as $\nu \rightarrow \infty$. Let us start by inspecting the univariate density function of the Student's t-distribution which is defined

$$f_t(x; \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}.$$

Taking the first fraction, we can find the limit as $\nu \rightarrow \infty$. We will use the Stirling approximation for the gamma function which states

$$\Gamma(z) \approx \sqrt{\frac{2\pi}{z}} \left(\frac{z}{e}\right)^z.$$

So, for $\Gamma(\frac{\nu+1}{2})$ and $\Gamma(\frac{\nu}{2})$, we get

$$\Gamma\left(\frac{\nu+1}{2}\right) \approx \sqrt{\frac{4\pi}{\nu+1}} \left(\frac{\nu+1}{2e}\right)^{\frac{\nu+1}{2}}, \quad \Gamma\left(\frac{\nu}{2}\right) \approx \sqrt{\frac{4\pi}{\nu}} \left(\frac{\nu}{2e}\right)^{\frac{\nu}{2}}.$$

Substituting in these expressions gives us

$$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} = \frac{\sqrt{\frac{4\pi}{\nu+1}} \left(\frac{\nu+1}{2e}\right)^{\frac{\nu+1}{2}}}{\sqrt{\pi\nu} \sqrt{\frac{4\pi}{\nu}} \left(\frac{\nu}{2e}\right)^{\frac{\nu}{2}}}.$$

We can simplify this expression into the following

$$\frac{1}{\sqrt{\pi\nu}} \cdot \frac{\sqrt{\frac{4\pi}{\nu+1}}}{\sqrt{\frac{4\pi}{\nu}}} \cdot \left(\frac{\nu+1}{2e}\right)^{\frac{1}{2}} \left[\left(1 + \frac{1}{\nu}\right)^{\nu}\right]^{\frac{1}{2}}.$$

First note that for large values of ν , $(1 + \frac{1}{\nu})^{\nu} \approx e$. Secondly, for large values of ν , $\nu+1 \approx \nu$ as the 1 become negligible. Therefore, for large ν , we get

$$\frac{\sqrt{\frac{\nu}{2}}}{\sqrt{\pi\nu}} = \frac{1}{\sqrt{2\pi}}.$$

Next, we look at the bracket of the density for the t-distribution and take $\nu \rightarrow \infty$.

$$\lim_{\nu \rightarrow \infty} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} = \lim_{\nu \rightarrow \infty} \left[\left(1 + \frac{x^2}{\nu}\right)^{\nu}\right]^{-\frac{1}{2}} = e^{-\frac{x^2}{2}}.$$

So, putting the first and second part together we get

$$\lim_{\nu \rightarrow \infty} f_t(x; \nu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = f_{Gauss}(x).$$

This is exactly the density of the standard Gaussian density. We now need to show that the cumulative distribution function of the t-distribution converges to the standard Gaussian cumulative distribution. Let F_t be the CDF of the t-distribution

$$\lim_{\nu \rightarrow \infty} F_t(x; \nu) = \lim_{\nu \rightarrow \infty} \int_{-\infty}^x f_t(x; \nu) dx.$$

By the Dominated Convergence Theorem, we have

$$\lim_{\nu \rightarrow \infty} \int_{-\infty}^z f_t(x; \nu) dx = \int_{-\infty}^z \lim_{\nu \rightarrow \infty} f_t(x; \nu) dx = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \Phi(z).$$

This can be extended to d -dimensions. The t-copula is of the form

$$C_{\nu, \rho}^t(\mathbf{u}) = t_{\nu}^d(t_{\nu}^{-1}(u_1), \dots, t_{\nu}^{-1}(u_d)).$$

So, we get

$$\lim_{\nu \rightarrow \infty} C_{\nu, \rho}^t = \Phi_{\Sigma}^d(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)) = C_{\rho}^G.$$

This example suggests that if we take large values for the degrees of freedom, we can get "more Gaussian" behaviour from the copula, a smaller value gives us fat tails like a true t-copula, increasing our flexibility in modelling.

2.7.3 Comparing Archimedean Copulas

The Frank, Clayton and Gumbel copulas all belong to the Archimedean family of copulas which are formed by a generator ϕ . They are each specialised for a certain type of tail dependence.

- Clayton copula - only lower tail dependence ($\lambda_L > 0, \lambda_U = 0$)
- Gumbel copula - only upper tail dependence ($\lambda_L = 0, \lambda_U > 0$)
- Frank copula - only lower tail dependence ($\lambda_L = 0, \lambda_U = 0$)

This gives us a good idea of what each copula could be used for. Clayton copulas are useful for extreme "low" events, Gumbel copulas for extreme "high" events and Frank copulas where the data is strongly correlated in the centre but independent in the extremes. When used for the correct problem, the specific copulas will perform well. However, if the data's tail dependence is not the same as the type in the copula, it will fail.

Chapter 3

Survival Analysis

3.1 Introduction

Survival analysis plays a central role in the analysis of time-to-event data, particularly in medical and clinical applications, where understanding the timing of outcomes is as important as the outcomes themselves [20].

We introduce the key concepts of survival analysis, which aim to model survival functions for time-to-event data. We define different types of events and censoring mechanisms, and then discuss a variety of methods for estimating survival functions. We then extend the univariate survival function to a multivariate setting by introducing joint survival functions defined through copulas, in order to model competing events and dependence structures.

This chapter is structured as follows. Section 3.2 defines survival data and common types of censoring encountered in such datasets. Section 3.3 describes the survival and hazard functions, as well as their relationship. Section 3.4 outlines the Cox proportional hazards model, while Section 3.5 introduces the accelerated failure time model. Section 3.6 discusses the differences between the Cox and AFT models and their appropriate use cases. Section 3.7 defines the Akaike and Bayesian information criteria and explains how they are used to compare models. Section 3.8 introduces the copula-based survival function, which is the main focus of this thesis, and Section 3.9 derives the tail dependence properties of a selection of copulas used in the simulation study.

3.2 Events, Censoring and Longitudinal Data

Survival analysis focuses on time-to-event outcomes, providing insight into survival patterns within populations. An event may correspond to death, disease progression, relapse, or the attainment of a clinically relevant threshold. The primary objective is to analyse the time until such an event occurs.

In practice, survival data are often incomplete because the event of interest may not be fully observed during the study period. In these situations, the survival times are said to be censored. The main types of censoring encountered in survival analysis are illustrated in Figure 3.1.

Right censoring: Right censoring occurs when the event has not taken place by the end of the observation period. In this case, we only know that the true event time exceeds the observed follow-up time (see Figure 3.1).

Interval censoring: Interval censoring arises when the event is known to have occurred within a specific time interval, but the exact time is unknown (Figure 3.1).

Left censoring: Left censoring refers to situations in which the event has already occurred before observation begins, implying that the true event time is less than the first observed time point (Figure 3.1).

In this dissertation, we restrict our attention to right censoring. That is, we assume that the event time may be unobserved only because the event has not occurred by the end of the follow-up period or because an individual leaves the study before experiencing the event.

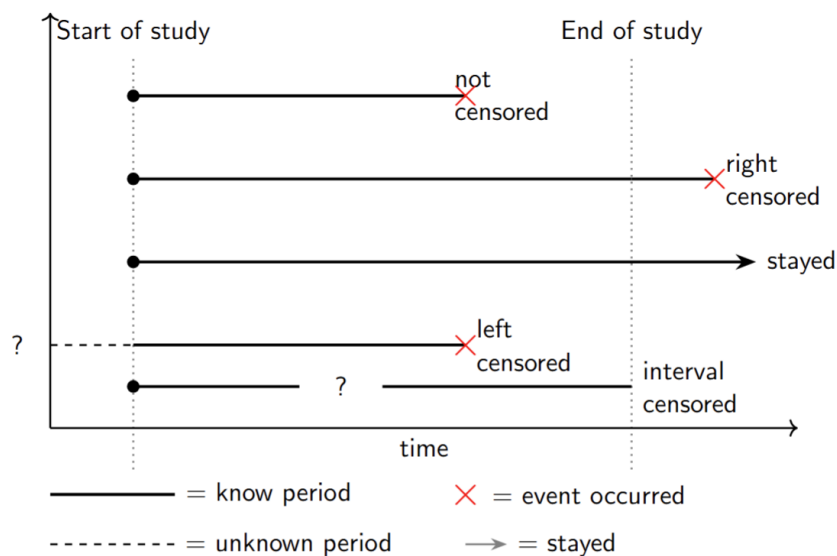


Figure 3.1: Illustration of different types of censoring mechanisms in survival analysis.

The data collected for analysis will be longitudinal data. Longitudinal data consist of repeated measurements collected from the same individual over time [13]. Such data are particularly important in survival analysis, as they provide dynamic information about a patient's condition and may help identify or predict the occurrence of an event.

These measurements may be continuous, such as biomarker levels or tumour size, or discrete, such as the appearance of a new lesion. For example, [23] discusses the Response Evaluation Criteria in Solid Tumours (RECIST), where repeated measurements of tumour size are treated as continuous outcomes, while the occurrence of new lesions represents

discrete outcomes. This provides a clear illustration of longitudinal data in a clinical setting, which will be explored further in Chapter 4.

3.3 Survival and Hazard Functions

Let T^* denote the non-negative random variable representing the true time to the event of interest. The distribution of T^* characterises the survival experience of an individual. In practice, T^* may not be fully observed due to censoring. Let C denote the censoring time. In this thesis, we focus on right censoring. Under right censoring, instead of observing T^* directly, we observe $T = \min(T^*, C)$, together with the censoring indicator $\delta = \mathbb{I}_{\{T^* \leq C\}}$, where

$$\delta = \begin{cases} 1, & \text{if the event time is observed,} \\ 0, & \text{if the observation is right censored.} \end{cases}$$

Thus, the observed data for each individual consist of the pair (T, δ) .

Definition 3.3.1. The survival probability of a patient at time $t \geq 0$ is the probability that the patient survives beyond time t . Formally, the survival function of the true event time T^* is defined as

$$S(t) = \mathbb{P}(T^* > t).$$

If F denotes the cumulative distribution function of T^* , then

$$S(t) = \mathbb{P}(T > t) = 1 - \mathbb{P}(T \leq t) = 1 - F(t).$$

Definition 3.3.2. The hazard function, $h(t)$, represents the instantaneous risk of the event occurring at time t , given that the individual has survived up to that time. Let $f(t)$ denote the probability density function of the true event time T^* , and $S(t)$ the corresponding survival function given by (3.3.1). Then the hazard function is defined as

$$h(t) = \frac{f(t)}{S(t)} = -\frac{d}{dt} \log S(t).$$

The cumulative hazard function is defined as

$$H(t) = \int_0^t h(s) ds,$$

which satisfies the relationship

$$S(t) = \exp(-H(t)).$$

The Exponential, Gamma and Weibull distributions are commonly used to model survival rates. Figure 3.2 shows these survival curves for different parameters, illustrating how changes in the distributional parameters affect the hazard behaviour over time, including constant, increasing, and decreasing hazard rates, and consequently the overall shape and steepness of the survival function.

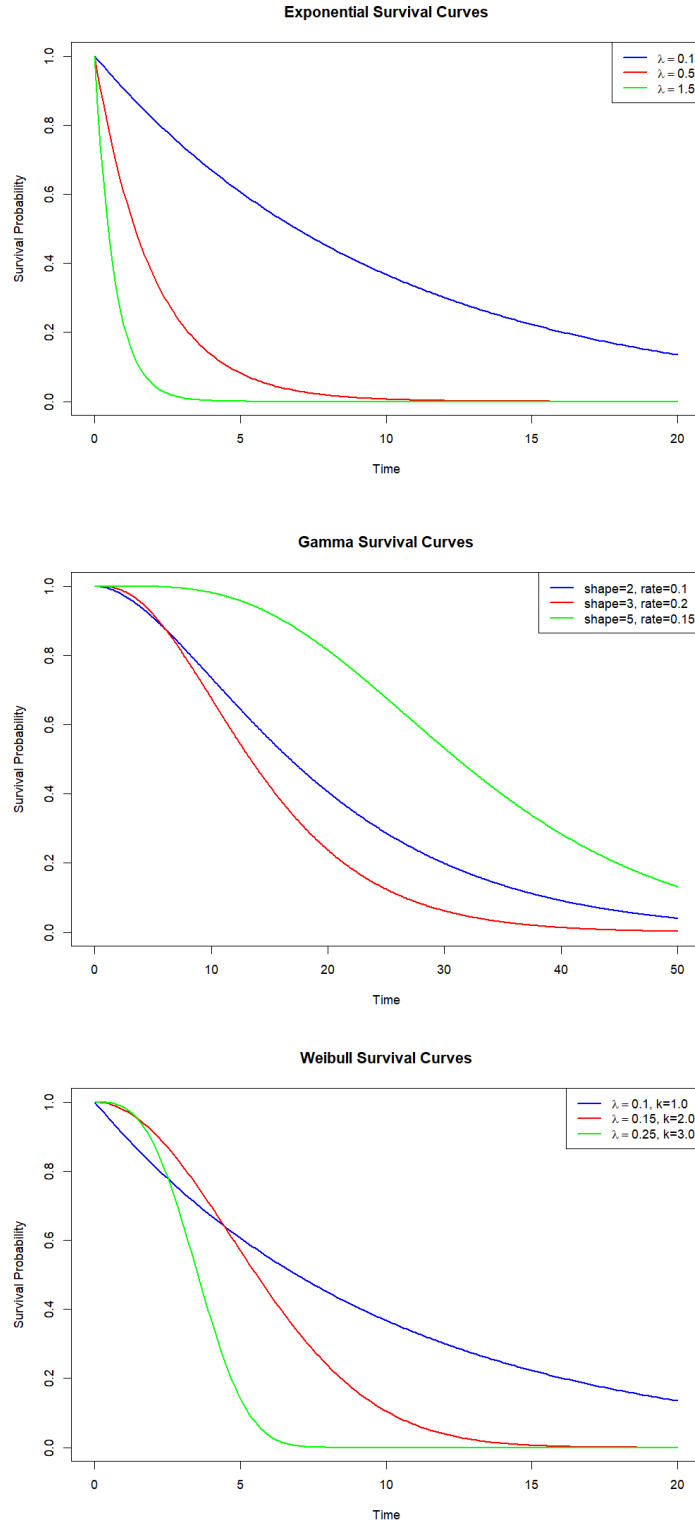


Figure 3.2: Survival functions of the Exponential, Gamma, and Weibull distributions for varying parameter values.

In practice, real data often do not follow a well-specified parametric distribution, making it difficult to directly estimate the survival and hazard functions. A common non-

parametric approach for estimating the survival function is the Kaplan-Meier estimator, introduced by Kaplan and Meier [17] in 1958.

Definition 3.3.3. Let our survival data set contain n patients, each with discrete event at time t which takes values in $\{t_1, \dots, t_T\}$. Define d_m as the number of events at time t_m and r_m as the number of patients at risk at the time t_m . The Kaplan-Meier estimator is defined

$$\hat{S}_{KM}(t) = \prod_{t_m \leq t} \left(1 - \frac{d_m}{r_m}\right).$$

It is sensible to consider the variance of our estimator which can be found via the delta method [6]. The delta method uses a Taylor series expansion to approximate the variance of the KM estimate. In [10], we can see how the variance is obtained. The variance of the KM estimate is given by

$$\text{Var}(\hat{S}_{KM}(t)) = (\hat{S}_{KM}(t))^2 \sigma(t)^2,$$

where $\sigma(t) = \prod_{t_m \leq t} \left(\frac{d_m}{r_m(r_m - d_m)}\right)$.

The 95% confidence interval is given by

$$[\hat{S}_{KM}(t) - z_{(1-\frac{\alpha}{2})}\sigma(t)\hat{S}_{KM}(t), \hat{S}_{KM}(t) + z_{(1-\frac{\alpha}{2})}\sigma(t)\hat{S}_{KM}(t)],$$

where $z_{1-\frac{\alpha}{2}}$ is the $1 - \frac{\alpha}{2}$ quantile of the standard normal distribution.

To derive the confidence interval we first need to compute the variance of the estimator through the delta method. The following proposition outlines the delta method [10].

Proposition 3.3.4. Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables such that

$$\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function. Then

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2 g'(\mu)^2).$$

Remark 3.3.5. The MLE follows a normal distribution asymptotically so we can approximate the variance of the function of the MLE using the delta method.

Let $\hat{h}_m = \frac{d_m}{r_m}$ be the MLE of the Kaplan-Meier estimate where d_m has a binomial distribution $d_m \sim \text{Bin}(r_m, h_m)$. Also, the log of the estimator is

$$\begin{aligned} \log(\hat{S}_{KM}(t)) &= \log \prod_{t_m \leq t} \left(1 - \frac{d_m}{r_m}\right) \\ &= \sum_{t_m \leq t} \log \left(1 - \frac{d_m}{r_m}\right). \end{aligned}$$

So we have the limiting distribution of $\hat{h}_m$ by the Central Limit Theorem given by

$$\sqrt{r_m}(\hat{h}_m - h_m) \xrightarrow{d} N(0, h_m(1 - h_m)).$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = \log(x)$ which is continuously differentiable. Then, by Proposition 3.3.4, the limiting distribution of $\log\left(1 - \frac{d_m}{r_m}\right)$ is given by

$$\sqrt{r_m}(\log(1 - \hat{h}_m) - \log(1 - h_m)) \xrightarrow{d} N\left(0, \frac{h_m}{1 - h_m}\right).$$

We can now find the variance of $\log(\hat{S}_{KM}(t))$.

$$\text{Var}(\log(\hat{S}_{KM}(t))) = \text{Var}\left(\sum_{t_m \leq t} \log(1 - \hat{h}_m)\right).$$

By the linearity of the variance and letting $\sigma(t)^2 = \text{Var}(\log(\hat{S}_{KM}(t)))$

$$\begin{aligned} \sigma(t)^2 &= \text{Var}(\log(\hat{S}_{KM}(t))) \\ &= \sum_{t_m \leq t} \text{Var}(\log(1 - \hat{h}_m)) \\ &= \sum_{t_m \leq t} \frac{h_m}{r_m(1 - h_m)} \\ &= \sum_{t_m \leq t} \frac{d_m}{r_m(r_m - d_m)}. \end{aligned}$$

To obtain the variance of $\hat{S}_{KM}(t)$, we apply the delta method again with a continuously differentiable $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = e^x$. So by Proposition 3.3.4

$$\text{Var}(\exp(\log(\hat{S}_{KM}(t)))) = \sigma(t)^2 [\hat{S}_{KM}(t)]^2.$$

Simplifying the left hand side of the equation, we get

$$\text{Var}(\hat{S}_{KM}(t)) = \sigma(t)^2 [\hat{S}_{KM}(t)]^2.$$

This is Greenwood's Formula for the variance of the Kaplan-Meier estimator. We can use this to construct confidence intervals for the estimator. The $100(1 - \alpha)\%$ confidence interval for $\hat{S}_{KM}(t)$ is given by

$$[\hat{S}_{KM}(t) - z_{(1-\alpha/2)}\sigma(t)\hat{S}_{KM}(t), \hat{S}_{KM}(t) + z_{(1-\alpha/2)}\sigma(t)\hat{S}_{KM}(t)],$$

where $z_{(1-\alpha/2)}$ is the $1 - \alpha/2$ percentile of the standard normal distribution.

Example 3.3.6. In this example, consider Exercise 3.2 in [36] but estimate $\hat{S}_{KM}$ instead of the hazard function. First, simulate $n = 1000$ observations from a Weibull distribution with failure rate $1/100$ and shape parameter 1.5 . Assume that censored times follow a Weibull distribution with rate $1/50$ and same shape parameter 1.5 . Fit a Kaplan-Meier estimator to the data and compare with the true survival curve.

In Figure 3.3, we have computed the KM estimate of 1000 samples from a Weibull distribution and its confidence intervals and compared it to the true survival curve of a Weibull distribution.

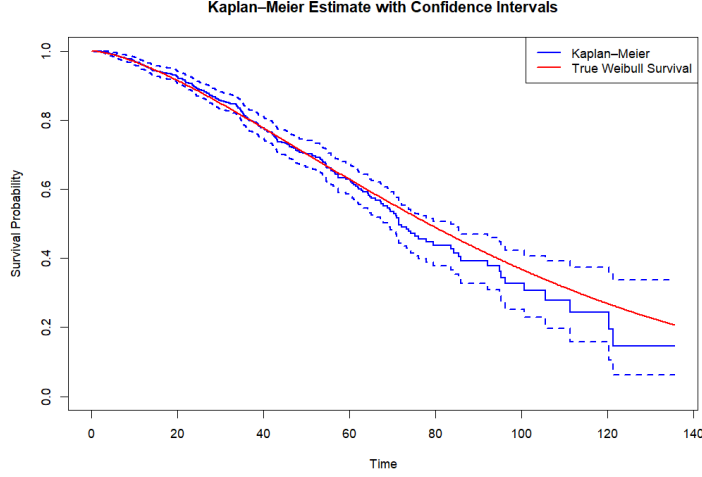


Figure 3.3: Kaplan–Meier estimate for Weibull-distributed survival data.

As we can see, the KM estimate is a good approximation of the true curve, especially at the start of the study. However, As more failures occur and the number of observations available to the Kaplan-Meier falls so the estimate becomes less accurate.

3.4 Cox Proportional Hazards (PH) Model

The Cox proportional hazards model is a regression framework based on the proportional hazards assumption, which states that the effects of covariates $\mathbf{x}$ (such as age, treatment group, and sex) remain constant over the time period of interest.

Definition 3.4.1. Let $\mathbf{x}$ be a vector of covariates, $\boldsymbol{\beta}$ a vector of regression coefficients, and $h_0(t)$ the baseline hazard function. The Cox proportional hazards model defines the hazard function for an individual as

$$h(t | \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}).$$

In this model, all individuals share the same baseline hazard function $h_0(t)$, while the term $\exp(\mathbf{x}^\top \boldsymbol{\beta})$ scales the hazard according to the individual's covariates.

Analogous to the Kaplan-Meier estimator for the survival function, we can also estimate the baseline hazard function $h_0(t)$. A commonly used estimator for this purpose is the Breslow estimator [4].

Definition 3.4.2. Let $\mathcal{R}_i$ denote the risk set just prior to T_i , $\mathbf{x}_m$ the covariate vector for patient m , and $\hat{\boldsymbol{\beta}}$ the maximum partial likelihood estimate of the regression coefficients in the Cox model.

Breslow's estimator of the baseline cumulative hazard function [37] is

$$\hat{H}_0(t) = \sum_{T_i \leq t} \frac{1}{\sum_{m \in \mathcal{R}_i} \exp(\mathbf{x}_m^\top \hat{\boldsymbol{\beta}})},$$

where the sum is over observed event times T_i (uncensored). The corresponding baseline survival function is then estimated as

$$\hat{S}_0(t) = \exp(-\hat{H}_0(t)).$$

Finally, the estimated survival function for an individual with covariates $\mathbf{x}$ is

$$\hat{S}(t | \mathbf{x}) = \hat{S}_0(t)^{\exp(\mathbf{x}^\top \hat{\boldsymbol{\beta}})}.$$

Example 3.4.3. We consider Exercise 5.3 from [36], in which the goal is to fit a Cox proportional hazards model to the **veteran** dataset available in the R package **survival**. This dataset contains information on veterans diagnosed with lung cancer. The covariates that will be included in our analysis are summarized in Table 3.1.

Table 3.1: Variable definitions for the **veteran** dataset.

Variable	Description
trt	Treatment: 1 = standard, 2 = test
celltype	Cell type: 1 = squamous, 2 = smallcell, 3 = adeno, 4 = large
time	Survival time
status	Censoring status
karno	Karnofsky performance score (100 = good)
diagtime	Months from diagnosis to randomisation
age	Age in years
prior	Prior therapy: 0 = no, 10 = yes

Our objective is to quantify the effect of each covariate on the hazard rate. Using the **survival** package in R, we can fit a Cox proportional hazards model to this dataset and estimate the regression coefficients associated with the covariates.

Table 3.2: Estimated hazard ratios and 95% confidence intervals from the Cox proportional hazards model.

Variable	Hazard Ratio (e^{β})	Lower Bound	Upper Bound
trt	1.343	0.896	2.013
celltype (smallcell)	2.368	1.390	4.033
celltype (adeno)	3.307	1.834	5.963
celltype (large)	1.494	0.859	2.599
karno	0.968	0.958	0.978
age	0.991	0.973	1.009
prior	1.007	0.967	1.049

The confidence intervals indicate statistical significance: if the interval includes 1, the effect is not statistically significant. For this Cox regression, cell types ‘smallcell’ and ‘adeno’ are statistically significant and have a hazard ratio greater than 1, so have a much higher risk of death compared to other cell types. The other statistically significant covariate ‘karno’, has a hazard ratio less than 1, so a 1-unit increase in ‘karno’ score results in an approximately 3% decrease in risk of death.

Now, we can use these coefficients to find the baseline hazard function. The function `basehaz` in R uses the Breslow estimate to compute an estimate for the hazard function. We get the graph of the estimate in Figure 3.4.

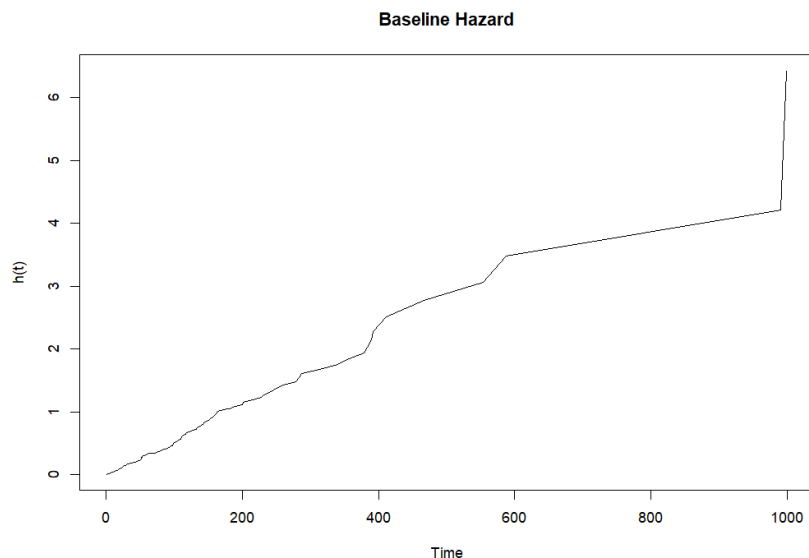


Figure 3.4: Breslow estimate of the baseline hazard function for the `veteran` dataset.

From the relationship between the hazard and survival functions, we can find the survival function of two patients with different covariates.

The covariates listed in Table 3.3 correspond to the survival curves shown in Figure 3.5. Similar curves can be generated for any combination of covariates, allowing the estimation of the survival function for individual patients.

Table 3.3: Covariate profiles for new patients used in Cox PH model predictions.

Patient	trt	celltype	karno	age	prior
New Patient 1	1	adeno	78	54	10
New Patient 2	1	smallcell	45	73	0

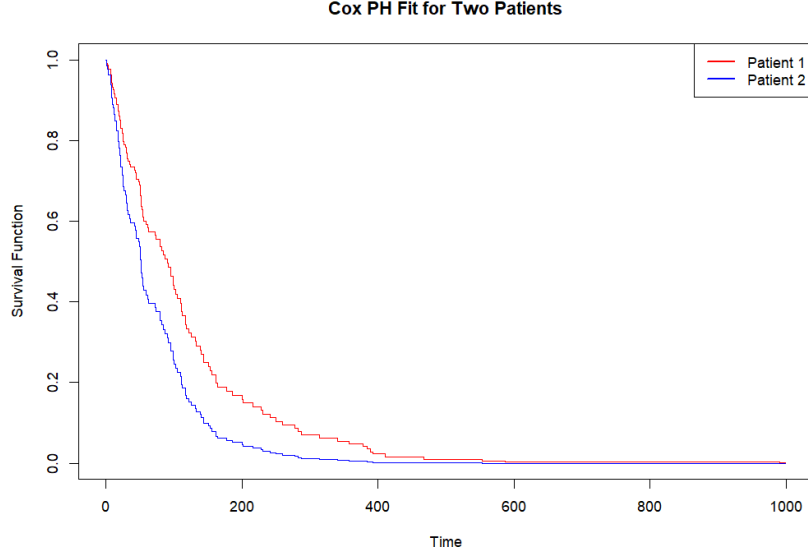


Figure 3.5: Estimated survival functions for two patients based on a Cox proportional hazards model.

3.5 Accelerated Failure Time (AFT) Models

In survival analysis, it is common to encounter situations where the proportional hazards assumption does not hold; that is, the effects of the covariates on the hazard function may change over time. An alternative approach is the accelerated failure time model [9], in which the effect of covariates is multiplicative on the event time rather than on the hazard. The AFT model is typically specified using known parametric distributions for the survival time, such as the Weibull, Gamma, Log-normal, and Log-logistic distributions [30].

Definition 3.5.1. Let $\mathbf{x}$ be a vector of covariates for a patient, $\boldsymbol{\beta}$ the corresponding vector of regression coefficients, and ϵ an error term with a specified distribution. The AFT model [9] is defined as

$$\log(T) = \mathbf{x}^\top \boldsymbol{\beta} + \epsilon.$$

From this log-linear formulation, we can derive the survival and hazard functions. Exponentiating both sides gives

$$T = \exp(\mathbf{x}^\top \boldsymbol{\beta} + \epsilon) = \exp(\mathbf{x}^\top \boldsymbol{\beta}) \exp(\epsilon).$$

Let $\epsilon_T = \exp(\epsilon)$. Then the survival function of T is

$$S(t) = \mathbb{P}(T \geq t) = \mathbb{P}(\epsilon_T \exp(\mathbf{x}^\top \boldsymbol{\beta}) \geq t) = \mathbb{P}\left(\epsilon_T \geq \frac{t}{\exp(\mathbf{x}^\top \boldsymbol{\beta})}\right).$$

If there are no covariates ($\mathbf{x} = \mathbf{0}$), the survival function reduces to

$$S_\epsilon(t) = \mathbb{P}(\epsilon_T \geq t).$$

Hence, for any covariate vector $\mathbf{x}$, the survival function can be expressed as

$$S(t \mid \mathbf{x}) = S_\epsilon \left(\frac{t}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} \right).$$

The corresponding hazard function is

$$h(t) = \frac{f(t)}{S(t)} = -\frac{S'(t)}{S(t)},$$

where f is the probability density function of T . Differentiating $S(t)$ with respect to t gives

$$S'(t) = \frac{d}{dt} S_\epsilon \left(\frac{t}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} \right) = \frac{1}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} S'_\epsilon \left(\frac{t}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} \right).$$

Thus, the hazard function for the AFT model is

$$h(t \mid \mathbf{x}) = -\frac{S'(t)}{S(t)} = \frac{1}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} h_\epsilon \left(\frac{t}{\exp(\mathbf{x}^\top \boldsymbol{\beta})} \right),$$

where $h_\epsilon(t) = -S'_\epsilon(t)/S_\epsilon(t)$ is the baseline hazard function of the error term ϵ_T .

Example 3.5.2. Suppose the survival times follow a Weibull distribution with scale parameter $\lambda > 0$ and shape parameter $k > 0$. To formulate this as an AFT model, we first determine the distribution of the error term.

Let $U \sim \text{Uniform}(0, 1)$, and recall that for any continuous random variable T ,

$$T = F_T^{-1}(U).$$

For the Weibull distribution, this gives

$$T = \lambda(-\log U)^{1/k}.$$

Taking the logarithm of both sides, we obtain

$$\log(T) = \log(\lambda) - \frac{1}{k} \log(-\log U).$$

The term $-\log(-\log U)$ follows a standard Gumbel distribution with location $\mu = 0$ and scale $\beta = 1$. Denoting $\epsilon \sim \text{Gumbel}(0, 1)$, we can write

$$\log(T) = \log(\lambda) + \frac{1}{k} \epsilon,$$

which is in the form of an accelerated failure time model. Hence, for a Weibull AFT model, the error term ϵ must be Gumbel-distributed.

The survival function of a standard Gumbel error term is

$$S_\epsilon(\epsilon) = \exp(-\exp(\epsilon)).$$

Using the AFT formulation $\log(T) = \mathbf{X}^\top \boldsymbol{\beta} + \epsilon$, we substitute $\epsilon = \log(T) - \mathbf{X}^\top \boldsymbol{\beta}$ to get the individual survival function:

$$S_i(t) = S_\epsilon(\log t - \mathbf{X}^\top \boldsymbol{\beta}) = \exp(-\exp(\log t - \mathbf{X}^\top \boldsymbol{\beta})) = \exp(-t \exp(-\mathbf{X}^\top \boldsymbol{\beta})),$$

where the last equality follows from properties of the exponential function. The corresponding hazard function is then

$$h_i(t) = -\frac{d}{dt} \log S_i(t) = \exp(-\mathbf{X}^\top \boldsymbol{\beta}) t^0 = \exp(-\mathbf{X}^\top \boldsymbol{\beta}),$$

which is proportional to the effect of the covariates in the AFT model.

Example 3.5.3. Let us return to Example 3.4.3, which uses the veterans dataset. We will fit a Weibull AFT model to these data. Using R, we can estimate the regression coefficients for the covariates, which are summarized in Table 3.4.

Similarly to the Cox proportional hazards model, we can use these covariates to estimate the survival function for individual patients. Using the covariate profiles shown in Table 3.3, the resulting survival curves for Patient 1 and Patient 2 are presented in Figure 3.6.

Table 3.4: Estimated hazard ratios and 95% confidence intervals from the Weibull accelerated failure time (AFT) model.

Variable	Hazard Rate (e^β)	Lower Bound	Upper Bound
trt	1.280	1.897	0.864
celltype (smallcell)	2.442	1.466	4.068
celltype (adeno)	3.389	1.967	5.839
celltype (large)	1.536	0.898	2.628
karno	0.968	0.958	0.978
age	0.993	0.976	1.011
prior	1.005	0.966	1.046

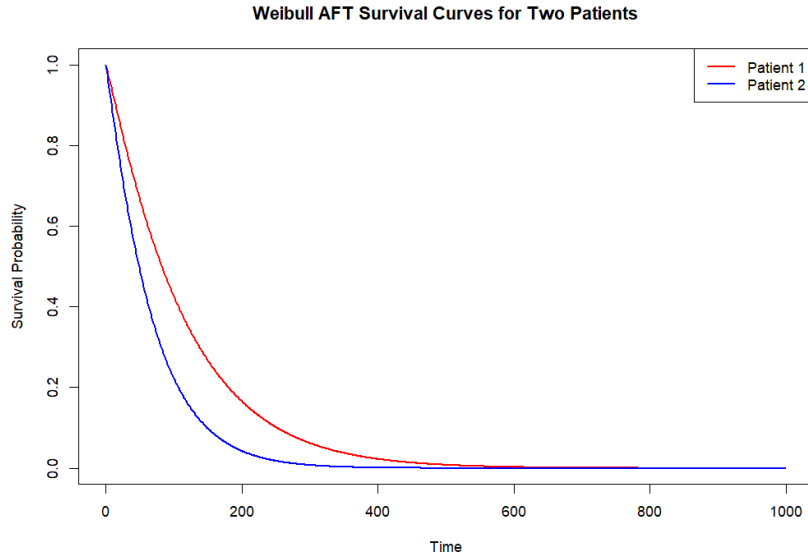


Figure 3.6: Estimated survival functions for two patients based on a Weibull accelerated failure time model.

3.6 Cox PH versus AFT Models

In survival analysis, selecting an appropriate model depends on the underlying assumptions and study objectives. The Cox proportional hazards model relies on the proportional hazards assumption, which states that the hazard ratio between any two individuals remains constant over time [8]. This feature is especially useful in clinical trials, as it allows inference on covariate effects without specifying the baseline hazard function. The proportional hazards assumption can be assessed using formal tests such as the Schoenfeld residual test [32] or graphical methods like log-minus-log survival plots [20].

A key advantage of the Cox PH model is its flexibility: it does not require a parametric form for the baseline hazard. However, when proportional hazards are violated—such as when treatment effects vary over time—the accelerated failure time model may be more appropriate. In the AFT framework, covariates act multiplicatively on survival time rather than on the hazard, effectively stretching or compressing the survival curve. This approach is particularly useful when the focus is on predicting actual survival times.

The AFT model requires specifying a parametric distribution for the survival times, such as Weibull, Gamma, Log-normal, or Log-logistic. Notably, the Weibull distribution is compatible with both modelling approaches: it can be formulated as a proportional hazards model or as an AFT model [7]. This duality allows analysts to select the most appropriate interpretation while maintaining consistency in parameter estimates.

In summary, the Cox PH model is preferred for its minimal distributional assumptions and flexibility, while the AFT model is advantageous when proportional hazards are violated or when a parametric survival distribution is known or assumed. Careful evaluation of the proportional hazards assumption should guide the choice between these models.

3.7 Comparing Models

While visual inspection of survival curves and model estimates can provide intuition on model behaviour, statistical criteria are essential for formally comparing models and selecting the most appropriate one. Two widely used criteria for this purpose are the Bayesian Information Criterion and the Akaike Information Criterion.

3.7.1 Bayesian Information Criterion (BIC)

Definition 3.7.1. Let k denote the number of parameters in the model, n the number of observations, and L the maximized likelihood. The Bayesian Information Criterion is defined as [24]:

$$BIC = -2\log(L) + k\log(n).$$

When comparing models, the model with the lower BIC is generally preferred. To compare two models with k_1 and k_2 parameters, we define

$$\Delta_{12} = BIC(k_1) - BIC(k_2).$$

According to [24], the magnitude of Δ_{12} provides evidence for model selection. Specifically, when $0 \leq \Delta_{12} \leq 2$, there is weak or negligible evidence to prefer one model over the other, whereas if $\Delta_{12} > 2$, there is evidence in favor of the model with the lower BIC.

3.7.2 Akaike Information Criterion (AIC)

Definition 3.7.2. Let k denote the number of parameters and L the maximized likelihood. The Akaike Information Criterion is defined as [29]:

$$AIC = 2k - 2\log(L).$$

Similar to the BIC, the model with the lower AIC is preferred.

Remark 3.7.3. Sometimes the AIC and BIC may disagree. In this work, we choose the model based on the BIC. The rationale is that AIC penalizes only the number of parameters k , whereas BIC penalizes both the number of parameters k and the number of observations n , taking into account the sample size. This often leads to a more conservative model selection when sample size is large.

Example 3.7.4. In Example 3.4.3, we fitted a Weibull AFT model to the veteran dataset. However, there are several alternative distributions that can be used in an AFT model, and a different choice may provide a better fit to the data. To compare model fit formally, we use the BIC, the AIC, and the log-likelihood. Table 3.5 summarizes these metrics for four different parametric AFT models.

As shown in Table 3.5, the Log-logistic distribution has the lowest AIC and BIC among the four candidate models. Therefore, when estimating a patient's survival function, the Log-logistic AFT model should be preferred.

Table 3.5: Comparison of parametric AFT models on the veteran dataset using log-likelihood, number of parameters (k), sample size (n), AIC, and BIC. Lower AIC or BIC indicates a better model fit.

Model	LogLik	k	n	AIC	BIC
Log-normal	-714.643	36	137	1501.286	1606.405
Weibull	-715.553	36	137	1503.106	1608.225
Log-logistic	-711.963	36	137	1495.926	1601.046
Gamma	-714.865	36	137	1501.730	1606.850

3.8 Survival Functions and Copulas

3.8.1 Joint Survival Functions

Subjects who are at risk of several types of events but can experience at most one are said to be subject to competing risks [28]. In our simulation study, we follow the RECIST

framework [1] to define the events a cancer patient may experience with respect to their tumours. The events considered are: (i) tumor growth beyond a certain threshold, and (ii) the appearance of a new lesion. These events are competing because a patient can experience only one of them before censoring occurs.

For univariate survival data, the traditional approach is the Cox model, which is well-studied and widely used. However, when multiple events are possible, a multivariate approach is required.

In our simulation, we focus on the time to tumour progression beyond a threshold and the time to a new lesion. Define, for the i -th subject ($i = 1, \dots, n$):

$$T_i^{(D)} := \text{time to new lesion}, \quad T_i^{(Y)} := \text{time to tumor progression}.$$

Consider the first event to occur. This is commonly referred to as progression-free survival (PFS) [35].

Definition 3.8.1. The progression-free survival time T for events T_D and T_Y is defined as

$$T = \min\{T_D, T_Y\}.$$

Definition 3.8.2. The joint survival function of T_D and T_Y is defined as

$$S_{T_D, T_Y}(t_D, t_Y) = \mathbb{P}(T_D > t_D, T_Y > t_Y).$$

In the univariate case, the marginal survival functions are

$$S_D(t_D) = \mathbb{P}(T_D > t_D) = S_{T_D, T_Y}(t_D, 0), \quad S_Y(t_Y) = \mathbb{P}(T_Y > t_Y) = S_{T_D, T_Y}(0, t_Y),$$

which can be estimated using the univariate methods described previously.

To model the progression-free survival curve for a patient, we consider

$$S_{PFS}(t) = \mathbb{P}(\min\{T_D, T_Y\} > t) = \mathbb{P}(T_D > t, T_Y > t) = S_{T_D, T_Y}(t, t).$$

3.8.2 Survival Copulas

In the previous section, we defined the joint survival function. Suppose we consider the univariate survival functions of T_D and T_Y as the marginal distributions for a survival copula $\tilde{C}$ [19]. Then, the joint survival function can be expressed as

$$S_{T_D, T_Y}(t_D, t_Y) = \tilde{C}(S_D(t_D), S_Y(t_Y)),$$

where the survival copula $\hat{C}$ is related to the standard copula C by

$$\tilde{C}(u, v) = u + v - 1 + C(1 - u, 1 - v).$$

To see why this holds, consider Sklar's Theorem [34]. Let F_D and F_Y denote the cumulative distribution functions of T_D and T_Y , respectively. The joint CDF is

$$H(t_D, t_Y) = C(F_D(t_D), F_Y(t_Y)).$$

The joint survival function is then

$$\begin{aligned}
S_{T_D, T_Y}(t_D, t_Y) &= \mathbb{P}(T_D > t_D, T_Y > t_Y) \\
&= \mathbb{P}(T_D > t_D) + \mathbb{P}(T_Y > t_Y) - \mathbb{P}(\{T_D > t_D\} \cup \{T_Y > t_Y\}) \\
&= S_D(t_D) + S_Y(t_Y) - (1 - \mathbb{P}(T_D \leq t_D, T_Y \leq t_Y)) \\
&= S_D(t_D) + S_Y(t_Y) - 1 + H(t_D, t_Y) \\
&= S_D(t_D) + S_Y(t_Y) - 1 + C(F_D(t_D), F_Y(t_Y)) \\
&= S_D(t_D) + S_Y(t_Y) - 1 + C(1 - S_D(t_D), 1 - S_Y(t_Y)).
\end{aligned}$$

Defining $u = S_D(t_D)$ and $v = S_Y(t_Y)$ gives the compact form of the survival copula:

$$S_{T_D, T_Y}(t_D, t_Y) = u + v - 1 + C(1 - u, 1 - v),$$

so that

$$\tilde{C}(u, v) = u + v - 1 + C(1 - u, 1 - v).$$

3.8.3 Inference for Archimedean Copulas

Archimedean copulas are widely used in medical studies because they can capture different types of dependence between clinical events. For example, the Clayton copula models “early-onset” dependence, the Gumbel copula models “late-onset” dependence, and the Frank copula models symmetric dependence without tail dependence. We now outline the derivation of key statistics and measures of dependence for these copulas. The log-likelihood $\ell(\theta)$ for a probability density function $f(\cdot)$ is defined as

$$\ell(\theta) = \sum_{i=1}^n \log(f(x_i; \theta)).$$

The score function, denoted by $U(\theta)$, is the derivative of the log-likelihood with respect to the parameter:

$$U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}.$$

To obtain the maximum likelihood estimate (MLE) of the copula parameter θ , we solve

$$U(\hat{\theta}) = 0,$$

for $\hat{\theta}$. The solution $\hat{\theta}$ is the MLE of the copula parameter.

Clayton Copula

The density of the Clayton copula is

$$\tilde{c}(u, v; \theta) = \frac{\partial^2 \tilde{C}}{\partial u \partial v} = \frac{\partial^2}{\partial u \partial v} \left(u + v - 1 + ((1 - u)^{-\theta} + (1 - v)^{-\theta} - 1)^{-1/\theta} \right),$$

which simplifies to

$$\tilde{c}(u, v; \theta) = (1 + \theta)((1 - u)(1 - v))^{-(\theta+1)} \left((1 - u)^{-\theta} + (1 - v)^{-\theta} - 1 \right)^{-1/\theta-2}.$$

Using the log-likelihood formula, we obtain

$$\begin{aligned} \ell(\theta) &= \frac{n}{1 + \theta} - (1 + \theta) \sum_{i=1}^n \log((1 - u_i)(1 - v_i)) \\ &\quad - \left(\frac{1}{\theta} + 2 \right) \sum_{i=1}^n \log \left((1 - u_i)^{-\theta} + (1 - v_i)^{-\theta} - 1 \right). \end{aligned}$$

Gumbel Copula

The density of the Gumbel copula is

$$\tilde{c}(u, v; \theta) = \frac{\partial^2 \tilde{C}}{\partial u \partial v} = \frac{\partial^2}{\partial u \partial v} \left(u + v - 1 + \exp \left(- \left[(-\log(1 - u))^\theta + (-\log(1 - v))^\theta \right]^{1/\theta} \right) \right).$$

Setting $\tilde{u} = -\log(1 - u)$ and $\tilde{v} = -\log(1 - v)$ for convenience, we can simplify to

$$\begin{aligned} \tilde{c}(u, v; \theta) &= \frac{1}{(1 - u)(1 - v)} (\tilde{u}\tilde{v})^{\theta-1} (\theta - 1 + (\tilde{u}^\theta + \tilde{v}^\theta)^{1/\theta}) \\ &\quad \times \frac{(\tilde{u}^\theta + \tilde{v}^\theta)^{1/\theta}}{(\tilde{u}^\theta + \tilde{v}^\theta)^2} C(1 - u, 1 - v). \end{aligned}$$

Applying the log-likelihood formula, we get

$$\begin{aligned} \ell(\theta) &= (\theta - 1) \log(\tilde{u}\tilde{v}) - \log((1 - u)(1 - v)) + \log \left(\theta - 1 + (\tilde{u}^\theta + \tilde{v}^\theta)^{1/\theta} \right) \\ &\quad - (\tilde{u}^\theta + \tilde{v}^\theta)^{1/\theta} + \left(\frac{1}{\theta} - 2 \right) \log(\tilde{u}^\theta + \tilde{v}^\theta). \end{aligned}$$

Frank Copula

Let $\tilde{u}_i = 1 - u_i$ and $\tilde{v}_i = 1 - v_i$. The density of the Frank copula is given by

$$\begin{aligned} \tilde{c}(u, v; \theta) &= \frac{\partial^2 \tilde{C}}{\partial u \partial v} = \frac{\partial^2}{\partial u \partial v} \left(u + v - 1 - \frac{1}{\theta} \log \left[1 + \frac{(e^{-\theta\tilde{u}} - 1)(e^{-\theta\tilde{v}} - 1)}{e^{-\theta} - 1} \right] \right) \\ &= \frac{\theta e^{-\theta(\tilde{u}+\tilde{v})}}{(1 - e^{-\theta}) \left(1 + \frac{(e^{-\theta\tilde{u}} - 1)(e^{-\theta\tilde{v}} - 1)}{e^{-\theta} - 1} \right)^2}. \end{aligned}$$

The corresponding log-likelihood function for a sample of size n is

$$\begin{aligned} \ell(\theta) &= n \log(\theta) - n \log(1 - e^{-\theta}) - \theta \sum_{i=1}^n (\tilde{u}_i + \tilde{v}_i) \\ &\quad - 2 \sum_{i=1}^n \log \left(1 + \frac{(e^{-\theta\tilde{u}_i} - 1)(e^{-\theta\tilde{v}_i} - 1)}{e^{-\theta} - 1} \right). \end{aligned}$$

Differentiating this log-likelihood and solving for $\hat{\theta}$ analytically is generally infeasible due to the complexity of the equations. In practice, numerical optimization methods, such as those implemented in R, are used to obtain the MLE of θ .

3.9 Dependence Measures for Copulas

3.9.1 Kendall's Tau

The relationship between Kendall's τ and the copula parameter can be derived explicitly for Archimedean copulas [27]. For this family, Kendall's tau has a convenient closed-form expression in terms of the copula generator. Let $\phi(t)$ denote the generator of the copula. Then,

$$\tau = 1 + 4 \int_0^1 \frac{\phi(t)}{\phi'(t)} dt.$$

This formula allows a direct connection between the copula parameter and the measure of concordance between the variables. In the following, we compute Kendall's τ for different members of the Archimedean family using their respective generators.

Clayton Copula

The Clayton copula has generator $\phi(t) = \frac{1}{\theta}(t^{-\theta} - 1)$. Using the formula for Kendall's tau, we have

$$\begin{aligned} \tau &= 1 + 4 \int_0^1 \frac{\frac{1}{\theta}(t^{-\theta} - 1)}{-t^{-\theta-1}} dt \\ &= 1 - \frac{4}{\theta} \left[\frac{t^2}{2} - \frac{1}{2+\theta} t^{2+\theta} \right]_0^1 \\ &= \frac{\theta}{2+\theta}. \end{aligned}$$

Gumbel Copula

The Gumbel copula has generator $\phi(t) = (-\log t)^\theta$. Then,

$$\begin{aligned} \tau &= 1 + 4 \int_0^1 \frac{(-\log t)^\theta}{-\frac{\theta}{t}(-\log t)^{\theta-1}} dt \\ &= 1 + \frac{4}{\theta} \left[\frac{1}{2} t^2 \log t - \frac{1}{4} t^2 \right]_0^1 \\ &= 1 - \frac{1}{\theta}. \end{aligned}$$

Frank Copula

The Frank copula has generator

$$\phi(t) = -\log\left(\frac{e^{-\theta t} - 1}{e^{-\theta} - 1}\right), \quad \phi'(t) = \frac{\theta e^{-\theta t}}{e^{-\theta t} - 1}.$$

The integral in Kendall's tau formula becomes

$$I = \int_0^1 \frac{\phi(t)}{\phi'(t)} dt = -\frac{1}{\theta} \int_0^1 \log\left(\frac{e^{-\theta t} - 1}{e^{-\theta} - 1}\right) \frac{e^{-\theta t} - 1}{e^{-\theta t}} dt.$$

Substituting $x = \theta t$ gives

$$I = -\frac{1}{\theta^2} \int_0^\theta \log\left(\frac{e^{-x} - 1}{e^{-\theta} - 1}\right) \frac{e^{-x} - 1}{e^{-x}} dx.$$

After integration by parts, this reduces to

$$I = -\frac{1}{\theta} \left[1 - \frac{1}{\theta} \int_0^\theta \frac{x}{e^x - 1} dx \right].$$

The integral is the Debye function $D_1(\theta)$:

$$D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{x}{e^x - 1} dx,$$

so that

$$I = \frac{1}{\theta} (D_1(\theta) - 1).$$

Hence, the Kendall tau for the Frank copula is

$$\tau = 1 + \frac{4}{\theta} (D_1(\theta) - 1).$$

These relationships provide simple formulas linking Kendall's tau to the copula parameter θ , which can be useful for estimating dependence in bivariate survival data.

3.9.2 Tail Dependence

Tail dependence coefficients, which describe the probability of simultaneous extreme events, are well-studied in Archimedean copulas [16]. The upper tail dependence coefficient of a copula C is defined as

$$\lambda_U = 2 - \lim_{t \rightarrow 1^-} \frac{1 - C(t, t)}{1 - t}.$$

We can derive the relationship between the tail dependence of a base copula C and its corresponding survival copula $\tilde{C}$. For the upper tail of the survival copula:

$$\begin{aligned}
\lambda_U^{\tilde{C}} &= 2 - \lim_{t \rightarrow 1^-} \frac{1 - \tilde{C}(t, t)}{1 - t} \\
&= \lim_{t \rightarrow 1^-} \frac{1 - 2t + \tilde{C}(t, t)}{1 - t} \\
&= \lim_{t \rightarrow 1^-} \frac{1 - 2t + (2t - 1 + C(1 - t, 1 - t))}{1 - t} \\
&= \lim_{t \rightarrow 1^-} \frac{C(1 - t, 1 - t)}{1 - t} \\
&= \lim_{s \rightarrow 0^+} \frac{C(s, s)}{s} \\
&= \lambda_L^C.
\end{aligned}$$

Similarly, the lower tail dependence of a copula is defined as

$$\lambda_L = \lim_{t \rightarrow 0^+} \frac{C(t, t)}{t}.$$

For the survival copula, the lower tail becomes

$$\begin{aligned}
\lambda_L^{\tilde{C}} &= \lim_{t \rightarrow 0^+} \frac{\tilde{C}(t, t)}{t} \\
&= \lim_{t \rightarrow 0^+} \frac{2t - 1 + C(1 - t, 1 - t)}{t} \\
&= \lim_{s \rightarrow 1^-} \frac{1 - 2s + C(s, s)}{1 - s} \\
&= \lambda_U^C.
\end{aligned}$$

Thus, for any copula C and its survival copula $\tilde{C}$:

$$\lambda_U^{\tilde{C}} = \lambda_L^C \quad \text{and} \quad \lambda_L^{\tilde{C}} = \lambda_U^C.$$

Table 3.6 summarizes these relationships for the Archimedean copulas considered.

Table 3.6: Tail dependence coefficients for Archimedean copulas and their corresponding survival copulas.

Copula	λ_L^C	λ_U^C	$\lambda_L^{\tilde{C}}$	$\lambda_U^{\tilde{C}}$
Clayton ($\theta > 0$)	$2^{-1/\theta}$	0	0	$2^{-1/\theta}$
Gumbel ($\theta \geq 1$)	0	$2 - 2^{1/\theta}$	$2 - 2^{1/\theta}$	0
Frank ($\theta \neq 0$)	0	0	0	0

Chapter 4

Copula Approach to RECIST Progression Framework

4.1 Introduction

In this chapter, we will describe the RECIST criteria which is the framework to classify progression of tumours in a patient. Then, we will define the Kaplan-Meier estimator in this scenario and outline how we will find the power of this method via the log-rank test. Finally, we will define the copula approach in respect to this framework, provide ways of assessing the power of this method and possible extensions. This chapter is organised as follows; Section 4.2 describes the RECIST criteria and the distributions and models the simulation data will be generated from, Section 4.3 defines the Kaplan-Meier estimator in the context of the RECIST criteria and Section 4.5 defines the copula approach in the context of the RECIST criteria.

4.2 Response Evaluation Criteria in Solid Tumours (RECIST)

We will review the Response Evaluation Criteria in Solid Tumours (RECIST) framework in detail, as it forms the focus of the simulation study in the next chapter. The purpose of this framework is to assess changes in tumours, identify new lesions and categorise the progression of a disease.

Firstly, tumour lesions are identified with a variety of methods such as CT scans, X-rays and clinical examinations and then categorised into measurable and non-measurable based on their longest diameter. The size of a lesion labelled measurable depends on the method of detection. These measurements are 10 mm by CT scan, 10 mm by physical caliper measurements and 20 mm by x-ray. Further details about measuring and detecting tumours can be found in [1].

The measurable tumours of interest are grouped into different responses; complete response (CR), partial response (PR), progressive disease (PD) and stable disease (SD). The simulation study will only focus on the progressive disease. However, the definitions of other categories can also be found in [1]. A progressive disease is defined as a tumour size greater than 20% of the minimum of all previous tumour sizes, or the appearance of one or more new lesions.

Let z_{it} be the tumour size of patient i at time t . Let z_{i0} be the baseline tumour size of patient i at time t . The log-tumour size ratio for patient i is given by

$$(Y_{i10}, \dots, Y_{iT0}) = \left(\log \left(\frac{z_{i1}}{z_{i0}} \right), \log \left(\frac{z_{i2}}{\min\{z_{i0}, z_{i1}\}} \right), \dots, \log \left(\frac{z_{iT}}{\min\{z_{i0}, \dots, z_{i(T-1)}\}} \right) \right),$$

where T is the number of follow-up times. The log tumour size ratios can be generated from a multivariate normal distribution. We can recover the original tumour sizes for patient i at time t with the following equation

$$z_{it} = \min\{z_{i0}, \dots, z_{i(T-1)}\} e^{y_{it}}.$$

Let $D_{it} \in \{0, 1\}$ where 0 indicates no new lesions and 1 indicates at least one new lesion. New lesion progression in a single arm trial can be modelled with a logistic regression model.

$$\text{Logit}\{\mathbb{P}(D_{it} = 1 | D_{i(t-1)} = 0, \dots, D_{i1} = 0, z_{i0}, \dots, z_{i(t-1)})\} = \alpha_t + \gamma_t z_{i(t-1)}$$

For a two arm comparative trial, new lesion progression can also be modelled with a logistic regression model, with an extra arm indicator term $R_i \in \{0, 1\}$ where the control arm is denoted 0 and the treatment arm is denoted 1.

$$\text{Logit}\{\mathbb{P}(D_{it} = 1 | D_{i(t-1)} = 0, \dots, D_{i1} = 0, z_{i0}, \dots, z_{i(t-1)})\} = \alpha_t + \beta_t R_i + \gamma_t z_{i(t-1)}.$$

Denote p_{it} as the probability of a new lesion appearing for the i -th patient at time t . Then we have

$$\text{Logit}\{p_{it}\} = \log \left(\frac{p_{it}}{1 - p_{it}} \right) = \alpha_t + \beta_t R_i + \gamma_t z_{i(t-1)}.$$

Rearranging to find p_{it} gives

$$p_{it} = \frac{1}{1 + e^{-(\alpha_t + \beta_t R_i + \gamma_t z_{i(t-1)})}}.$$

Remark 4.2.1. The regression coefficients α_t , β_t and γ_t have subscript t because they can be time-varying over the duration of the study. In Section 5.2, α_t , β_t and γ_t will remain constant in an individual scenario.

4.3 Kaplan-Meier Method

4.3.1 The Kaplan-Meier Estimator

The Kaplan-Meier was first proposed in [17] and [36] gives a clear method to apply it to RECIST in this section. To use the Kaplan-Meier approach, we must first define two

binary indicators that denote progression for each variable, patient and time. The i -th patient at time t has indicators

$$E_{it}^{(D)} = D_{it} \in \{0, 1\}, \quad E_{it}^{(Y)} = \mathbb{I}(z_{it} > 1.2 \times \min\{z_{i0}, \dots, z_{i(T-1)}\}),$$

where $\mathbb{I}$ is the indicator function. The event time T_i is the first time either a new lesion appears or the tumour size crosses the threshold. Let $\mathcal{T} = \{t_1, \dots, t_T\}$ be the set of follow-up times. The observed time X_i is given by

$$X_i = \min \left\{ \{t \in \mathcal{T} : E_{it}^{(D)} = 1 \text{ or } E_{it}^{(Y)} = 1\} \cup \{t_T\} \right\},$$

where t_T is the final follow up time. Finally, we need a status indicator which is defined

$$\delta_i = \begin{cases} 1 & \text{if } \exists t \in \mathcal{T} : E_{it}^{(D)} = 1 \text{ or } E_{it}^{(Y)} = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Now, the Kaplan-Meier estimate is

$$\hat{S}_{KM}(t) = \prod_{t_m \leq t} \left(1 - \frac{d_m}{r_m} \right),$$

where d_m is the number of patients experiencing their first new lesion or tumour growth at time t_m and r_m is the number of patients who are at risk just before t_m .

4.3.2 The Log-Rank Test

The log-rank test gives us a way to evaluate if two survival curves are statistically equivalent [21]. Consider two different groups 1 and 2. We define

$$d_{km} := \text{number of events in group } k \text{ at time } t_m,$$

$$r_{km} := \text{number of subjects at risk in group } k \text{ at time } t_m.$$

The expected number of events for group k and time t_m is defined

$$E_k = \sum_{m=1}^T r_{km} \frac{d_{1m} + d_{2m}}{r_{1m} + r_{2m}}.$$

The summed observed events in group k is denoted O_k . The test statistic is given by

$$\chi_1^2 = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2}.$$

The null hypothesis being tested is that there is no overall difference between the two survival curves. Under the null hypothesis, the log-rank statistic is approximately chi-square with one degree of freedom.

4.3.3 Power Calculations for Kaplan-Meier Estimators

In order to assess how effective the Kaplan-Meier estimator is at detecting a treatment effect, we need to compute the power. We will use the log-rank test in the power calculations.

The power calculations are defined in the following steps. Firstly, we run M simulations of the comparative trial and calculate the p-value of the log-rank test in each simulation. We record each simulation that has a p-value less than that α as an indicator function $\mathbb{I}(p^{(m)} < \alpha)$ where m is the simulation number. Then, the estimated power of the Kaplan-Meier estimate is given by

$$\text{Power} = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(p^{(m)} < \alpha),$$

where α is the significance level, usually $\alpha = 0.05$.

4.4 The Augmented-Binary Method

4.4.1 The Augmented-Binary Estimate

The augmented-binary (Aug-bin) method is an alternative way to consider two competing endpoints in survival analysis and was proposed in [22]. We consider a continuous endpoint Y , which is the log tumour size ratio generated from a multivariate normal, and a binary endpoint D which is the new lesion indicator whose probability is generated from a logistic regression model. Both of these variables were formally defined in Section 4.2.

For the augmented binary method, we want to define progression in terms of the generating distribution. The actual tumour sizes can be recovered from

$$z_{it} = \min\{z_{i0}, \dots, z_{i(T-1)}\} e^{y_{it}}.$$

Tumour size progression occurs when

$$z_{it} > c \cdot \min\{z_{i0}, \dots, z_{i(T-1)}\},$$

where c is the progression threshold. Substituting the tumour size formula, we get

$$\begin{aligned} \min\{z_{i0}, \dots, z_{i(T-1)}\} e^{y_{it}} &> c \cdot \min\{z_{i0}, \dots, z_{i(T-1)}\} \\ e^{y_{it}} &> c \\ y_{it} &> \log(c). \end{aligned}$$

So, the probability of a tumour progressing is

$$\mathbb{P}(\text{"Tumour Progression"}) = \mathbb{P}(y_{it} > \log(c)).$$

Therefore, the survival function for Y is

$$\begin{aligned} S_Y(t) &= \mathbb{P}(\text{"No Tumour Progression"}) \\ &= 1 - \mathbb{P}(\text{"Tumour Progression"}) \\ &= 1 - \mathbb{P}(y_{it} > \log(c)). \\ &= \mathbb{P}(y_{it} < \log(c)). \end{aligned}$$

We can write this survival function in terms of an integral,

$$S_Y(t) = \int_{-\infty}^{\log(c)} \cdots \int_{-\infty}^{\log(c)} f_{Y_1, \dots, Y_T}(y_1, \dots, y_t; \theta) dy_1 \cdots dy_t,$$

where θ is the vector of parameters of the model.

For the new lesion variable D , the logistic regression model of the conditional probabilities in Section 4.2 will be used. We have

$$S_D(t) = \mathbb{P}(D_1 = 0) \mathbb{P}(D_2 = 0 | D_1 = 0) \cdots \mathbb{P}(D_t = 0 | D_1 = \cdots = D_{t-1} = 0).$$

The augmented binary method aggregates the two survival functions to form a survival estimate of patient i at time t , given by

$$S_i(t|\theta) = \int_{\Omega^{t+k}} \mathbb{P}(D_1 = \cdots = D_t = 0) \cdot f_{\mathbf{Y}_i}(\mathbf{y}_i; \theta) d\mathbf{y}_i,$$

where $\Omega = (-\infty, \log(c))$, $k = \sum_{j=1}^t (j-1)$ and $\mathbf{y}_i = (y_{i1}, \dots, y_{it})$ are the log tumour size rations of patient i .

We will use two different methods for estimating the variance and confidence intervals of the augmented-binary method. Firstly, the bootstrap method will be used and percentile confidence intervals will be drawn from the samples. This approach to the augmented-binary estimate is denoted Aug-boot. The bootstrap method [6] is full described in Section 4.5.2 and follows. The delta method will be used as a second method to obtain the variance and confidence intervals due to it's quicker computational time in the power comparisons. This will be denoted Aug-delta. The delta method is described in [10].

4.4.2 Power Calculations

For the augmented-binary method, we wish to test if the hazard rates are statistically different with null hypothesis $H_0 : h(t|R=0) = h(t|R=1)$, $\forall t \leq \tau$, where τ is the maximum time where both groups have at least one subject still at risk and R is the arm indicator (control= 1, treatment= 1).

To test for differences between curves, we will use a pooled Z statistic defined by

$$Z(\tau) = \sum_{i=1}^{\tau} \left(h(t|R=1) - \frac{h(t|R=0) + h(t|R=1)}{2} \right),$$

where under the null hypothesis, $\frac{Z^2}{Var(Z)}$ has a chi-squared distribution with one degree of freedom. We reject H_0 if

$$\frac{Z^2}{Var(Z)} > \chi_{1,(1-\alpha)}^2,$$

where α is the significance level. $Var(Z)$ will be computed with the delta method for speed of computation.

To calculate the power, we will run M iterations of the comparative trial and find the proportion of trials where a difference is detected. We compute the sum

$$\text{Power} = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(p^{(m)} < \alpha),$$

where $p^{(m)}$ is the p value of the m -th trial and α is the significance level, usually $\alpha = 0.05$.

4.5 Copula Approach

4.5.1 The Copula-based Survival Estimate

To use the copula-defined joint survival function, we will begin by define the events individually. $E_{it}^{(D)}$ and $E_{it}^{(Y)}$ are defined in the same way as Section 4.3. Define the observed times $X_i^{(D)}$ and $X_i^{(Y)}$ by

$$X_i^{(D)} = \min\{\{t \in \mathcal{T} : E_{it}^{(D)} = 1\} \cup \{t_T\}\}, X_i^{(Y)} = \min\{\{t \in \mathcal{T} : E_{it}^{(Y)} = 1\} \cup \{t_T\}\}.$$

where t_T is the final follow-up time. The status indicator for each event is defined

$$\delta_i^{(D)} = \begin{cases} 1 & \text{if } \exists t \in \mathcal{T} : E_{it}^{(D)} = 1 \\ 0 & \text{otherwise} \end{cases}, \delta_i^{(Y)} = \begin{cases} 1 & \text{if } \exists t \in \mathcal{T} : E_{it}^{(Y)} = 1 \\ 0 & \text{otherwise} \end{cases}.$$

Then we can estimate the survival function of both variables individually with the Kaplan-Meier estimator given by

$$\hat{S}_{KM}^{(D)}(t) = \prod_{t_m \leq t} \left(1 - \frac{d_m^{(D)}}{r_m^{(D)}}\right), \hat{S}_{KM}^{(Y)}(t) = \prod_{t_m \leq t} \left(1 - \frac{d_m^{(Y)}}{r_m^{(Y)}}\right).$$

We could also consider a Cox model outlined in Section 3.4. Taking previous tumour size $z_{i(t-1)}$ and treatment group R_i as covariates. Then, the model has the form

$$h(t|R_i, z_{i(t-1)}) = h_0(t) \exp(\beta_t R_i + \gamma z_{i(t-1)}),$$

where $h_0(t)$ is the baseline hazard function which can be estimated. After estimating the baseline hazard function and the regression coefficients, we can use the relationship between the hazard and survival function to obtain

$$\hat{S}_{Cox}^{(D)}(t_D) = \exp(-\hat{h}^{(D)}(t_D|R_i, z_{i(t-1)})).$$

Consider the joint survival function defined by the survival copula

$$S_{T_D, T_Y}(t_D, t_Y) = \tilde{C}(S_D(t_D), S_Y(t_Y)),$$

where the survival copula $\tilde{C}$ is related to a standard copula by

$$\tilde{C}(u, v) = u + v - 1 + C(1 - u, 1 - v; \theta).$$

If we use the Kaplan-Meier estimate for the new lesion variable, we get

$$S_{T_D, T_Y}(t_D, t_Y) = \hat{S}_{KM}^{(D)}(t_D) + \hat{S}_{KM}^{(Y)}(t_Y) - 1 + C(1 - \hat{S}_{KM}^{(D)}(t_D), 1 - \hat{S}_{KM}^{(Y)}(t_Y); \hat{\theta}).$$

Instead, if we use the Cox estimate for the survival function, we get

$$S_{T_D, T_Y}(t_D, t_Y) = \hat{S}_{Cox}^{(D)}(t_D) + \hat{S}_{KM}^{(Y)}(t_Y) - 1 + C(1 - \hat{S}_{Cox}^{(D)}(t_D), 1 - \hat{S}_{KM}^{(Y)}(t_Y); \hat{\theta}).$$

Following on from Section 3.8.1, the estimate for the progression-free survival function at time t is given by

$$S_{PFS}(t) = S_{T_D, T_Y}(t, t) = \hat{S}_{KM}^{(D)}(t) + \hat{S}_{KM}^{(Y)}(t) - 1 + C(1 - \hat{S}_{KM}^{(D)}(t), 1 - \hat{S}_{KM}^{(Y)}(t); \hat{\theta}).$$

Note that the Cox model will not be used in further analysis.

4.5.2 The Bootstrap Method

The bootstrap method [6] is used to construct an empirical distribution of an estimated parameter. This is achieved by repeatedly resampling the observed data with replacement to generate bootstrap samples. The parameter of interest is then estimated for each sample, and the resulting estimates are used to approximate its sampling distribution, often visualised as a histogram.

In the case of the joint survival function, the delta method applied to the two-step estimation framework from [33] results in complex distributions where confidence intervals are hard to derive analytically. The bootstrap method offers an alternative way to obtain confidence intervals for the survival function.

Let B be the number of bootstrap samples and let n be the number of samples in the original data set. First, we draw n samples from the data with replacement to get the b -th bootstrap sample, $\mathcal{B}^b$, which is given by

$$\mathcal{B}^b = \{(T_i^{(D,b)}, T_i^{(Y,b)}, \delta_i^{(D,b)}, \delta_i^{(Y,b)})\}_{i=1}^n,$$

for $b = 1, \dots, B$. Then, we calculate the Kaplan-Meier estimates for variables D and Y for the b -th bootstrap sample. For ease of notation, we will drop the KM from the subscript and add (b) to indicate which bootstrap estimate this is. On $\mathcal{B}^b$, let $t_1^{(b)} < t_2^{(b)} \dots$ be the distinct observed event times for margin j where $j = D$ or Y . Then,

$$\hat{S}_D^{(b)} = \prod_{t_\ell < t} \left(1 - \frac{d_\ell^{(D,b)}}{r_\ell^{(D,b)}}\right), \quad \hat{S}_Y^{(b)} = \prod_{t_\ell < t} \left(1 - \frac{d_\ell^{(Y,b)}}{r_\ell^{(Y,b)}}\right).$$

Now, we substitute the pair of estimates $(\hat{S}_D^{(b)}(t), \hat{S}_Y^{(b)}(t))$ into the progression-free survival function defined by a copula C in Section 4.5

$$S_{PFS}^{(b)}(t) = \hat{S}_D^{(b)}(t) + \hat{S}_Y^{(b)}(t) - 1 + C(1 - \hat{S}_D^{(b)}(t), 1 - \hat{S}_Y^{(b)}(t); \theta^{(b)}).$$

We can estimate $\theta^{(b)}$ through maximum likelihood estimation to obtain $\hat{\theta}^{(b)}$. Once this is done, we have the b -th bootstrap estimate of $S_{PFS}(t)$ given by

$$\hat{S}_{PFS}^{(b)}(t) = \hat{S}_D^{(b)}(t) + \hat{S}_Y^{(b)}(t) - 1 + C(1 - \hat{S}_D^{(b)}(t), 1 - \hat{S}_Y^{(b)}(t); \hat{\theta}^{(b)}).$$

This method is repeated B times until we have B bootstrap estimates of the survival function. Each estimate is stored in the collection $\{\hat{S}_{PFS}^{(b)}(t)\}_{b=1}^B$. Once all samples have been estimated at each time point, we can compute summary statistics of the B bootstrap estimates, such as the mean and the $\alpha/2$ and $1 - \alpha/2$ quantiles, as well as construct a histogram. The quantiles will form the percentile confidence interval. As an alternative, we can construct a confidence interval via the normal approximation.

Constructing confidence intervals for an estimate through the normal approximation [11] requires the estimator to be asymptotically normal. For the estimator $\hat{S}_{PFS}(t)$ with standard error σ , the confidence interval is given by

$$[\hat{S}_{PFS}(t) - z_{1-\alpha/2}\hat{\sigma}, \hat{S}_{PFS}(t) + z_{1-\alpha/2}\hat{\sigma}],$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of the standard normal distribution. We can estimate σ via the method in [5].

4.5.3 Power Calculations for the Copula Estimate

We do not have a simple readily available test such as the log-rank test for the Kaplan–Meier estimate. Instead, we use the integrated absolute difference between survival curves to compare arms, adapting the approach of [2]. The null and alternative hypotheses are

$$H_0 : \hat{S}_{PFS}^{(1)}(t) = \hat{S}_{PFS}^{(2)}(t) \text{ for all } t \leq \tau, \quad H_1 : \hat{S}_{PFS}^{(1)}(t) \neq \hat{S}_{PFS}^{(2)}(t) \text{ for some } t \leq \tau,$$

where $\hat{S}_{PFS}^{(1)}$ and $\hat{S}_{PFS}^{(2)}$ are the copula-derived survival curves of the control and treatment arms, computed from arm-specific Kaplan–Meier marginals for tumour and lesion failure times glued by a fitted bivariate Gumbel copula via

$$\hat{S}_{PFS}(t) = \hat{S}_D(t) + \hat{S}_Y(t) - 1 + \hat{C}(1 - \hat{S}_D(t), 1 - \hat{S}_Y(t)).$$

The marginals $\hat{S}_D$ and $\hat{S}_Y$ are Kaplan–Meier estimates fitted separately to the tumour and lesion failure times in each arm; the copula $\hat{C}$ is a Gumbel bivariate copula with dependence parameter estimated by maximum likelihood on patients with both events observed.

We summarise the difference between arms by the integrated mean absolute difference

$$L_1 = \frac{1}{b} \sum_{t=a}^b |\hat{S}_{PFS}^{(2)}(t) - \hat{S}_{PFS}^{(1)}(t)|,$$

where a is the smallest pooled event time and b is the smallest of the two arms' maximum follow-up times. By construction $L_1 \geq 0$, with larger values indicating greater separation between arms.

Because the asymptotic distribution of L_1 under H_0 is not available in closed form, we obtain a p -value by a permutation procedure. Under H_0 the joint distribution of (tumour, lesion) outcomes is identical in both arms, so arm labels are exchangeable. We pool the $2n$ patients, draw B random permutations of the arm labels (without replacement), and on each permutation refit the copula in each pseudo-arm and recompute the statistic, denoted $L_1^{(b)}$. The one-sided p -value is

$$p = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(L_1^{(b)} \geq L_1).$$

To estimate the power of the copula-based test, we generate M independent simulated trials under H_1 and compute a p -value for each. The power is then estimated as

$$\text{Power} = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(p^{(m)} < \alpha),$$

where α is the chosen significance level, usually 0.05.

Chapter 5

Simulation Studies Data Analysis

5.1 Introduction

Chapter 4 outlined the stages involved in obtaining the Kaplan–Meier, augmented-binary and copula-based estimates, as well as the performance metrics used to assess them. This chapter presents the full framework for the simulation study.

Firstly, data are generated in R using the framework described in Section 4.2. For the Kaplan–Meier approach, the time of the first event (tumour progression or new lesion) is recorded. A Kaplan–Meier estimator is then fitted using a large simulated sample of 10^5 patients via the `survival` package to approximate the true survival function. In addition, an empirical estimate is obtained using a sample size of 150 patients, repeated over 1000 simulations to construct confidence intervals, from which coverage probabilities and average confidence interval widths are calculated.

The augmented-binary method uses the formula derived in Chapter 4 to estimate the survival function. Then, 1000 iterations of 200 bootstrap samples will be used to obtain confidence intervals and the delta method will be used in the power analysis for computational efficiency.

For the copula-based method, event times for each individual component (tumour progression and new lesion) are first obtained, and separate Kaplan–Meier estimates are computed for each marginal distribution. These are then combined using the copula formulation described in Section 4.5 to estimate the joint survival function. A bootstrap procedure with 200 resamples is repeated over 1000 iterations in order to evaluate coverage probability and average confidence interval width.

For the comparative study, several scenarios are considered to assess the power to detect differences between control and treatment groups. Finally, the proposed methods are applied to a real clinical cancer trial to evaluate their performance in a practical setting.

The aim of this chapter is to evaluate the proposed copula-based approach for survival data using both simulated and real datasets. The simulation framework focuses on tumour progression and the occurrence of new lesions, as described in the RECIST frame-

work [22]. All analyses are implemented in R, and a custom package, `tumourSim`, was developed to generate data, perform estimation, and compute summary statistics. The full documentation is available at <https://github.com/Calum1888/tumourSim>.

This chapter is organised as follows. Section 5.2 presents the parameters, variables, and scenarios used in the single-arm and comparative trials and Section 5.3 reports the results of these studies, including estimates, coverage probabilities, and confidence intervals.

5.2 Simulation Study

A single arm trial is a type of study where all participants are assigned to the same treatment group, and there is no parallel study with a different treatment to compare with. It is usually used by clinicians to assess outcomes such as safety or initial efficacy. In the case of our simulation study, we will use it to assess how effective the copula method is at predicting the survival rate compared to the Kaplan-Meier and augmented-binary methods.

For the single arm trial, we will assess the performance of the described methods through the coverage rate and average confidence interval width. The coverage rate is the proportion of confidence intervals that contain the true value. The average confidence width tells us how precise the confidence interval is so a smaller confidence interval width is preferred.

5.2.1 Single Arm Trial

For 150 patients, we will simulate data at 5 and 7 time points. A baseline tumour size, z_0 will be sampled from a uniform distribution, $z_0 \sim U(0, 1)$. Log tumour size ratio is sampled from a multivariate normal distribution $\mathbf{Y} \sim N_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where for 5 time points, we have mean $\boldsymbol{\mu}^{(5)} = (0.00, 0.036, 0.072, 0.108, 0.144)^T$ and covariance matrix

$$\boldsymbol{\Sigma}^{(5)} = \begin{bmatrix} 0.25 & 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.45 & 0.45 & 0.45 & 0.45 \\ 0.25 & 0.45 & 0.50 & 0.50 & 0.50 \\ 0.25 & 0.45 & 0.50 & 0.75 & 0.75 \\ 0.25 & 0.45 & 0.50 & 0.75 & 1.00 \end{bmatrix}.$$

For 7 time points, the mean is $\boldsymbol{\mu}^{(7)} = (-0.22, -0.29, -0.36, -0.36, -0.43, -0.51, -0.36)^T$ and the covariance matrix $\boldsymbol{\Sigma}^{(7)}$ has the form $\sigma_{ii} = 0.10$ and $\sigma_{ij} = 0.15$, $\forall i \neq j$.

Section 4.2 details the logistic regression of the probability of a new lesion $p_{D_{it}}$ given by

$$\text{Logit}(p_{D_{it}}) = \alpha_t + \gamma_t z_{i(t-1)},$$

where we disregard the arm indicator for the single arm trial. We set $(\alpha_t, \gamma_t) = (-2.5, 0.2)$ at all time points. Then, we will take 1000 iterations of 200 bootstrap samples to generate confidence intervals for $\hat{S}_{PFS}(t)$.

5.2.2 Comparative Trials

Comparative trials are used to compare two different groups. This could be a control group with no treatment compared to an actual medicine or two different treatments compared against each other. They can be used to determine which treatment is safer or more effective. In this simulation study, they will be used to test the power of detecting a difference between each treatment arm for each method.

In a comparative trial, we will simulated data for 200 patients in each arm at 5 time points where the patients are randomly allocated to the control ($R = 0$) and treatment arms ($R = 1$). Similarly to Section 5.2.1, the true survival rates of each arm are computed by a Kaplan-Meier estimate with 10^5 patients.

The mean of the treatment arm is given by $\boldsymbol{\mu} = (-0.2, -0.4, -0.56, -0.6, -0.65)$ and the covariance matrix is given by

$$\boldsymbol{\Sigma} = \begin{bmatrix} 0.05 & 0.05 & 0.05 & 0.05 & 0.05 \\ 0.05 & 0.10 & 0.10 & 0.10 & 0.10 \\ 0.05 & 0.10 & 0.14 & 0.14 & 0.14 \\ 0.05 & 0.10 & 0.14 & 0.16 & 0.16 \\ 0.05 & 0.10 & 0.14 & 0.16 & 0.18 \end{bmatrix}.$$

The control arm has mean $\boldsymbol{\mu} + \mathbf{a}$ where $\mathbf{a} = (0.1, \dots, 0.1)$ is a vector of 0.1s with the same number of entries as there is time points and the covariance is the same as the treatment arm.

Since a treatment arm is being incorporated, we will use the corresponding logistic regression model.

$$\text{Logit}(p_{D_{it}}) = \alpha_t + \beta_t R_i + \gamma_t z_{i(t-1)}$$

We will consider several scenarios to assess the power of the different methods.

Scenario 1: True null (Type I). This scenario sets $\beta_t = 0$ and removes all tumour-lesion dependence ($\gamma_{\text{ctrl}} = \gamma_{\text{trt}} = 0$). Treatment and control arms follow identical tumour trajectories, and lesion processes evolve independently. This provides a baseline for assessing empirical Type I error under complete independence.

Scenario 2: No gamma, small treatment effect. A mild treatment effect is introduced ($\beta_t = -0.2$) while maintaining $\gamma_{\text{ctrl}} = \gamma_{\text{trt}} = 0$. Lesions remain independent of tumour behaviour in both arms. This tests sensitivity to weak treatment effects when no multivariate structure is available to exploit.

Scenario 3: No gamma, medium treatment effect. A moderate effect ($\beta_t = -0.5$) is applied with no tumour-lesion dependence. This represents a typical non-null setting where survival curves diverge meaningfully but the lesion process provides no additional information.

Scenario 4: No gamma, large treatment effect. A strong treatment effect ($\beta_t = -0.8$) is combined with $\gamma_{\text{ctrl}} = \gamma_{\text{trt}} = 0$. This scenario checks whether methods behave as expected under large proportional hazards when lesion information is uninformative.

Scenario 5: Differing gamma, no treatment effect. Here $\beta_t = 0$, but tumour-lesion dependence differs sharply between arms ($\gamma_{\text{ctrl}} = 1.0$, $\gamma_{\text{trt}} = 0.2$). Although there is no treatment effect, the dependence structure is misaligned across arms. This tests robustness of methods to heterogeneity in multivariate dependence under the null.

Scenario 6: Differing gamma, small treatment effect. A moderate treatment effect ($\beta_t = -0.5$) is combined with asymmetric dependence ($\gamma_{\text{ctrl}} = 1.0$, $\gamma_{\text{trt}} = 0.2$). This scenario evaluates whether methods can detect treatment differences when the lesion process is more tightly coupled to tumour behaviour in the control arm than in the treatment arm.

Scenario 7: Differing gamma, medium treatment effect. A strong treatment effect ($\beta_t = -0.8$) is paired with the same asymmetric dependence structure. This tests whether copula-based methods gain efficiency when the control arm exhibits strong tumour-lesion coupling while the treatment arm does not.

Scenario 8: Heavy-tail null with dependence misalignment. A null scenario with no treatment effect ($\beta_t = 0$) and heavy-tailed multivariate t tumour distribution ($\nu = 4$). Dependence is set to $\gamma_{\text{ctrl}} = \gamma_{\text{trt}} = 0$, but the heavy tails induce occasional extreme tumour values. This scenario examines robustness of methods to non-Gaussian tumour dynamics and tail-driven variability.

Scenario 9: Heavy-tail H1 with dependence misalignment. A moderate treatment effect ($\beta_t = -0.5$) is combined with a heavy-tailed tumour distribution ($\nu = 4$) and asymmetric dependence ($\gamma_{\text{ctrl}} = 1.0$, $\gamma_{\text{trt}} = 0.2$). This tests whether copula methods can exploit both heavy-tailed structure and dependence asymmetry to improve efficiency under the alternative.

Scenario 10: Heavy-tail null with matched dependence. A null scenario ($\beta_t = 0$) with heavy-tailed tumour dynamics ($\nu = 4$) and matched dependence across arms ($\gamma_{\text{ctrl}} = \gamma_{\text{trt}} = 0.2$). This isolates the effect of heavy tails alone, without dependence asymmetry, to assess Type I error stability under non-Gaussian joint behaviour.

5.3 Simulation Results

5.3.1 Single Arm Results

Table 5.1 and Table 5.2 shows the true survival rate, estimated survival rate, coverage rate and average confidence interval width for three different Archimedean copulas; Clayton, Frank and Gumbel, across 5 and 7 time points respectively. Two different lengths of study were selected, 5 and 7 time points, to see if the proposed methods extends to longer studies whilst maintaining similar coverage rates and average confidence interval widths. Three different types of Archimedean copulas were chosen to capture different types of tail dependence for the possible trajectories of the trial. Clayton copulas have lower tail dependence which models late progression, Gumbel copulas have upper tail dependence to capture early progression and Frank copulas have no tail dependence. The true survival rates are generated by a Kaplan-Meier estimate of 10^5 patients. The

estimated values are generated from the `tumourSim` package using 1000 iterations of 200 bootstrap samples.

To choose the best copula to compare to the Kaplan-Meier and augmented-binary estimates, the coverage rate and average confidence interval width will be taken into consideration. The coverage rate informs us how often the method is correct and the average confidence interval width tells us how precise the method is.

In Table 5.1, the Gumbel copula is clearly the best performer. It has the smallest average confidence interval width, ranging from 0.189 to 0.095 and the second highest coverage rate, though all of the coverage rates for the Gumbel copula exceed the nominal 95% target. Furthermore, the point estimates of the survival rates for the Gumbel copula are extremely accurate, either no errors at time points 3 and 5 or small errors of 0.2%, 0.2% and 0.1% at time points 1, 2 and 4 respectively. The Clayton copula has marginally higher coverage rates compared to the Gumbel copula, apart from at time point 5, but has significantly larger average confidence interval widths across all time points, which could be inflating the coverage rate of the Clayton copula. The Clayton copula is slightly less accurate as well with errors ranging from 0.1% to 0.07%. Finally, the Frank copula performs poorly in this study, none of the coverage rates across the 5 time points reach the nominal 95% target and the point estimates of survival rate have larger errors, ranging from 1.6% to 4.4%.

Table 5.1: Comparison of estimated copula methods for 5 time points. Results are based on 1000 iterations of 200 bootstrap samples.

Time	True $S(t)$	Copula $S(t)$			Estimated Coverage Rate			Average CI Width		
		Clayton	Frank	Gumbel	Clayton	Frank	Gumbel	Clayton	Frank	Gumbel
1	0.587	0.597	0.571	0.589	0.977	0.947	0.965	0.804	0.214	0.189
2	0.329	0.336	0.293	0.331	0.975	0.902	0.959	0.781	0.220	0.178
3	0.213	0.218	0.169	0.213	0.973	0.858	0.958	0.681	0.189	0.153
4	0.115	0.119	0.079	0.114	0.971	0.827	0.960	0.277	0.131	0.118
5	0.063	0.067	0.039	0.063	0.983	0.846	0.992	0.147	0.090	0.095

In Table 5.2, the Gumbel copula was also the best performer. Similarly to the 5 time points study, the Gumbel copula had the smallest average confidence interval width with values from 0.137 to 0.062 and the second highest coverage rate which all reached or exceeded the nominal 95% target. The point estimates were also exact to the true rate in the cases of time points 4, 6 and 7, or had very small errors of 0.1% at time points 1, 2, 3 and 5. Also, following the trend of Table 5.1, the Clayton copula had the highest coverage rates but also the highest average confidence interval width, which ranged from 0.554 to 0.060, once again suggesting that the wider confidence intervals could be increasing the coverage rate. Finally, the Frank copula performed poorly again, with only 2 of the time points reaching the nominal 95% target. In the comparison with the Kaplan-Meier and augmented-binary methods, the Gumbel copula, which was clearly the best type of copula across 5 and 7 time points, will be used.

Table 5.2: Comparison of estimated copula methods for 7 time points. Results are based on 1000 iterations of 200 bootstrap samples.

Time	True $S(t)$	Copula $S(t)$			Estimated Coverage Rate			Average CI Width		
		Clayton	Frank	Gumbel	Clayton	Frank	Gumbel	Clayton	Frank	Gumbel
1	0.748	0.753	0.742	0.749	0.975	0.951	0.953	0.483	0.146	0.137
2	0.514	0.515	0.492	0.513	0.975	0.929	0.953	0.554	0.172	0.151
3	0.335	0.337	0.304	0.335	0.976	0.896	0.950	0.531	0.164	0.138
4	0.186	0.187	0.156	0.186	0.967	0.834	0.955	0.317	0.127	0.110
5	0.110	0.110	0.086	0.109	0.977	0.836	0.956	0.131	0.093	0.086
6	0.068	0.069	0.051	0.068	0.972	0.827	0.965	0.092	0.068	0.068
7	0.023	0.024	0.016	0.023	0.989	0.990	0.986	0.069	0.049	0.062

Table 5.3 and Table 5.4 show the true survival rates, estimated survival rates, coverage rate and average confidence interval width for the Kaplan-Meier estimate, the augmented-binary estimates and the Gumbel copula method. The results were computed from the custom R package `tumourSim`. The Kaplan-Meier estimator was chosen to compare the copula method to because it is one of the most widely used estimators in survival analysis and the go-to initial analysis that is performed on survival data. The augmented-binary method with bootstrap was also chosen as a comparison. This method saw increased precision compared to the Kaplan-Meier estimate in [22]. It is computed from 1000 iterations of 200 bootstrap samples. True survival rates were computed from a Kaplan-Meier estimate with 10^5 patients.

For five time points (Table 5.3), the point estimates are essentially identical across methods and closely match the true survival rate at every time point, indicating none of the methods are biased. The Kaplan-Meier estimator has the widest confidence intervals throughout. The copula method offers a modest reduction in CI width over Kaplan-Meier, ranging from 2.1% at time point 1 to 9.2% at time point 4, while the Aug-boot method achieves substantially larger reductions, with 36.3% at time point 1 to 62.1% at time point 5. Coverage tells a complementary story. Aug-boot achieves coverage of 92.5% to 94.5%, closest to the nominal 95% level. The Kaplan-Meier is conservatively over-covering at 94.9% to 98.0% and the copula sits between them at 95.8% to 99.2%. In summary, the Aug-boot is the most precise method and best calibrated to nominal coverage, the Kaplan-Meier is the most conservative, and the copula offers an intermediate trade-off.

Table 5.3: Comparison of estimated KM, Augbin and best performing copula method for 5 time points. KM results are based on 1000 iterations. Aug-boot and copula results are based on 1000 iterations of 200 bootstrap samples.

Time	True $S(t)$	Estimated $S(t)$			Estimated Coverage Rate			Average CI Width		
		KM	Aug-boot	Copula	KM	Aug-boot	Copula	KM	Aug-boot	Copula
1	0.587	0.587	0.589	0.589	0.960	0.943	0.965	0.193	0.123	0.189
2	0.329	0.330	0.330	0.331	0.949	0.945	0.959	0.186	0.109	0.178
3	0.213	0.213	0.213	0.213	0.949	0.931	0.958	0.163	0.088	0.153
4	0.115	0.114	0.114	0.114	0.961	0.936	0.960	0.130	0.060	0.118
5	0.063	0.063	0.062	0.063	0.980	0.925	0.992	0.103	0.039	0.095

For 7 time points (Table 5.4), the results follow a similar trend to Table 5.3. The point estimates across all time points for all the methods are identical to the true rate or only differ by 0.001 to 0.002. Again, the Kaplan-Meier has the widest confidence intervals across all time points apart from at time point 7 where the copula has a marginally wider confidence interval. The copula method does offer narrower confidence intervals everywhere else compared to the Kaplan-Meier with reductions ranging from 0.7% to 19.1%. Similarly to Table 5.3, the Aug-boot method has the narrowest confidence intervals and offers significant reductions in the confidence interval width compared to the Kaplan-Meier with the smallest reduction being 17.4% and the largest at 75.4%. The coverage rate also follows the trend at 5 time points. Aug-boot achieves coverage of 92.5% to 94.9%, which is closest to the nominal 95% level. Once again, the Kaplan-Meier is conservatively over-covering at 94.0% to 98.3% and the copula sits between them at 95.0% to 98.6%. In summary, the Aug-boot is the most precise method again and best calibrated to nominal coverage, the Kaplan-Meier is the most conservative, and the copula offers an intermediate trade-off.

Table 5.4: Comparison of estimated KM, Augbin and best performing copula method for 7 time points. KM results are based on 1000 iterations. Aug-boot and copula results are based on 1000 iterations of 200 bootstrap samples.

Time	True $S(t)$	Estimated $S(t)$			Estimated Coverage Rate			Average CI Width		
		KM	Aug-boot	Copula	KM	Aug-boot	Copula	KM	Aug-boot	Copula
1	0.748	0.750	0.748	0.749	0.940	0.930	0.953	0.138	0.114	0.137
2	0.514	0.514	0.513	0.513	0.956	0.949	0.953	0.160	0.121	0.151
3	0.335	0.335	0.335	0.335	0.949	0.943	0.950	0.152	0.102	0.138
4	0.186	0.185	0.186	0.186	0.953	0.939	0.955	0.126	0.071	0.110
5	0.110	0.108	0.111	0.109	0.951	0.927	0.956	0.102	0.048	0.086
6	0.068	0.067	0.069	0.068	0.965	0.925	0.965	0.084	0.033	0.068
7	0.023	0.022	0.023	0.023	0.983	0.930	0.986	0.057	0.014	0.062

The conclusions from Table 5.3 and Table 5.4 are very similar. This suggests all three methods are able to be extended from 5 to 7 time points.

Figure 5.1, Figure 5.2 and Figure 5.3 show the survival curves of the Kaplan-Meier, Aug-boot and copula methods for 5 time points. Both the estimated survival curve and true survival curve are plotted on each graph and they overlapped at essentially every time point which shows the accuracy of all the point estimates. The shaded regions are the average confidence widths. These figures highlight the extent to which the augmented-binary with bootstrap approach reduces the confidence interval width compared to the Kaplan-Meier and copula methods.

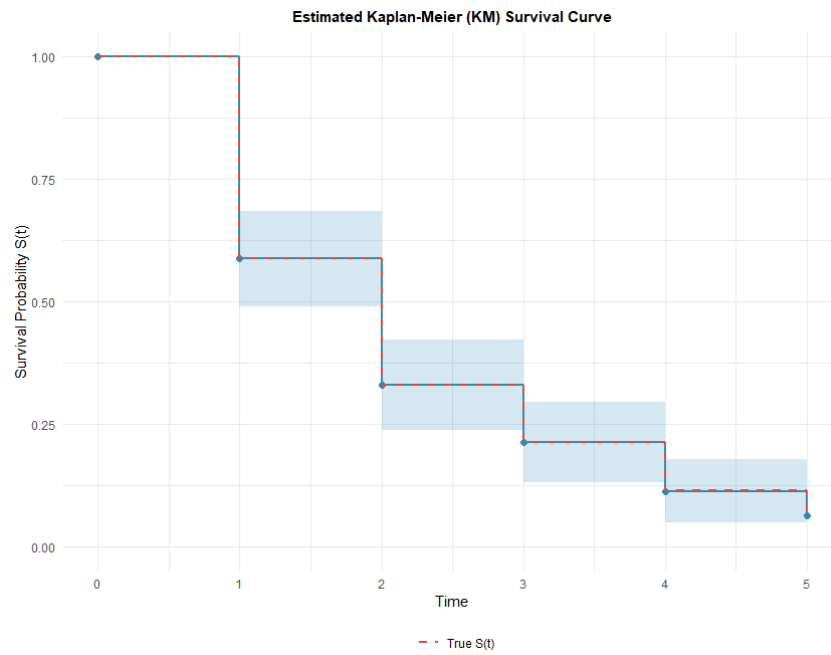


Figure 5.1: Estimated Kaplan-Meier survival curve for 5 time points. The shaded regions are confidence intervals computed from 1000 iterations.

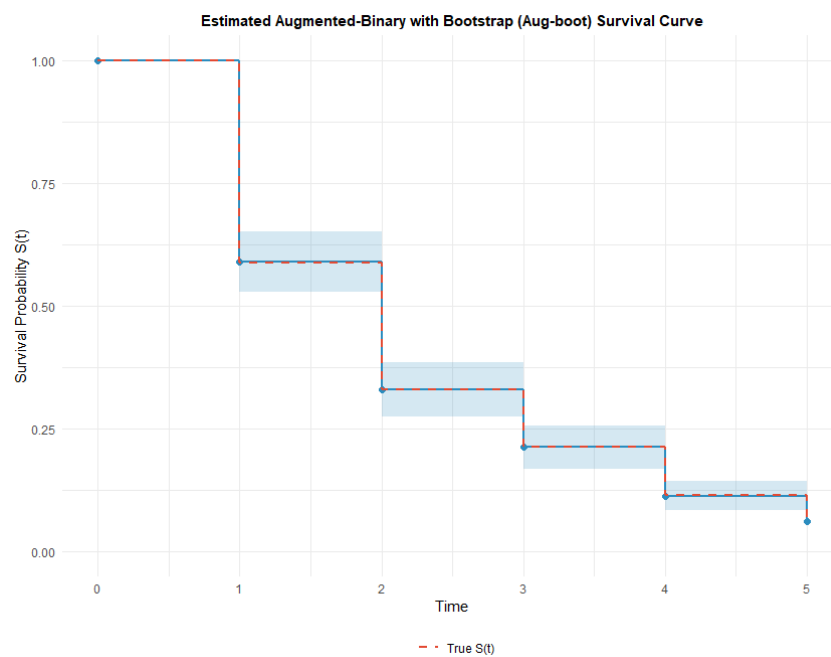


Figure 5.2: Estimated augmented-binary with bootstrap survival curve for 5 time points. Shaded regions are confidence intervals computed from 1000 iterations of 200 bootstrap samples.

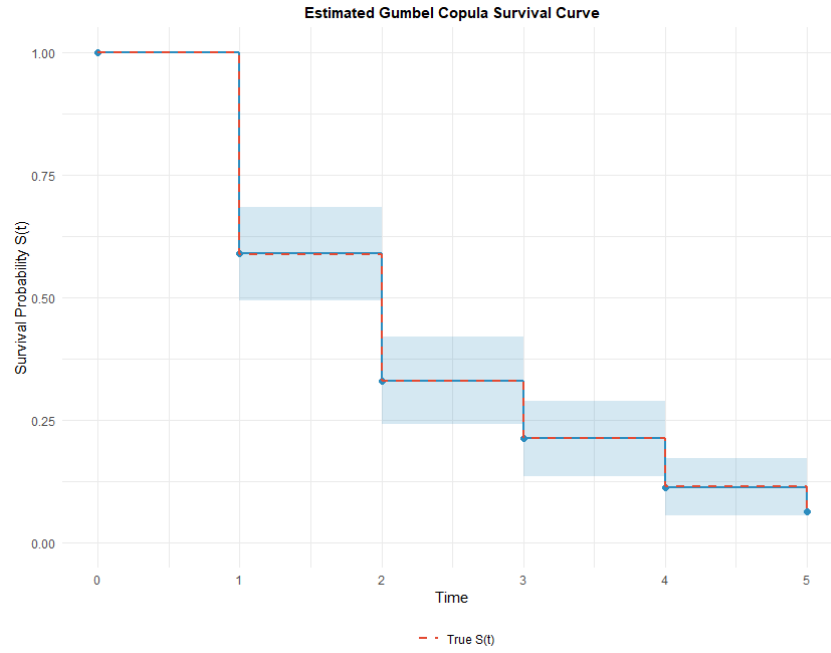


Figure 5.3: Estimated Gumbel copula survival curve for 5 time points. Shaded regions are confidence intervals computed from 1000 iterations of 200 bootstrap samples.

Figure 5.4, Figure 5.5 and Figure 5.6 show the survival curves of the Kaplan-Meier, Aug-boot and copula methods for 7 time points. They highlight the same observations made in the previous paragraph, that all methods have very accuracy point estimates and the augmented-binary with bootstrap method is extremely precise compared to the Kaplan-Meier and copula estimates.

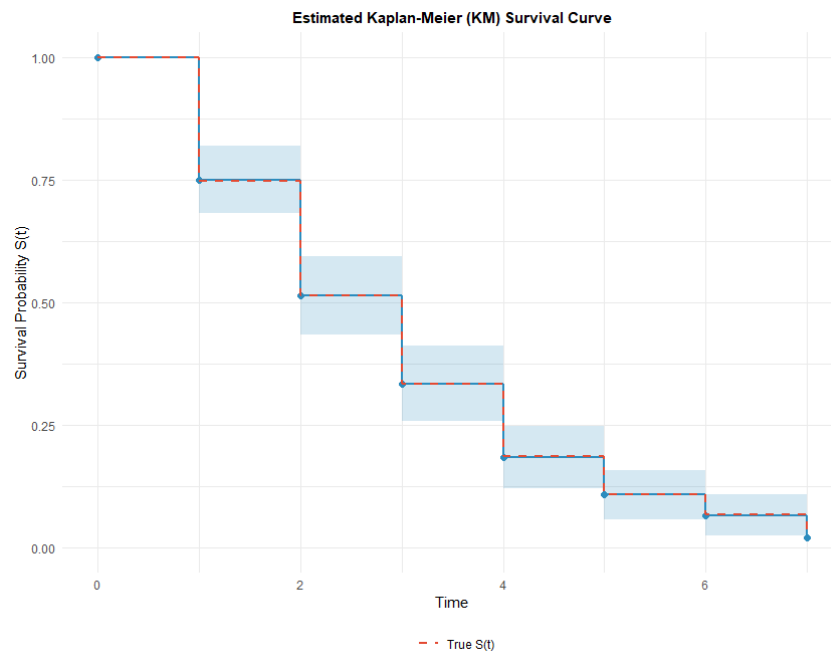


Figure 5.4: Estimated Kaplan-Meier survival curve for 7 time points. The shaded regions are confidence intervals computed from 1000 iterations.

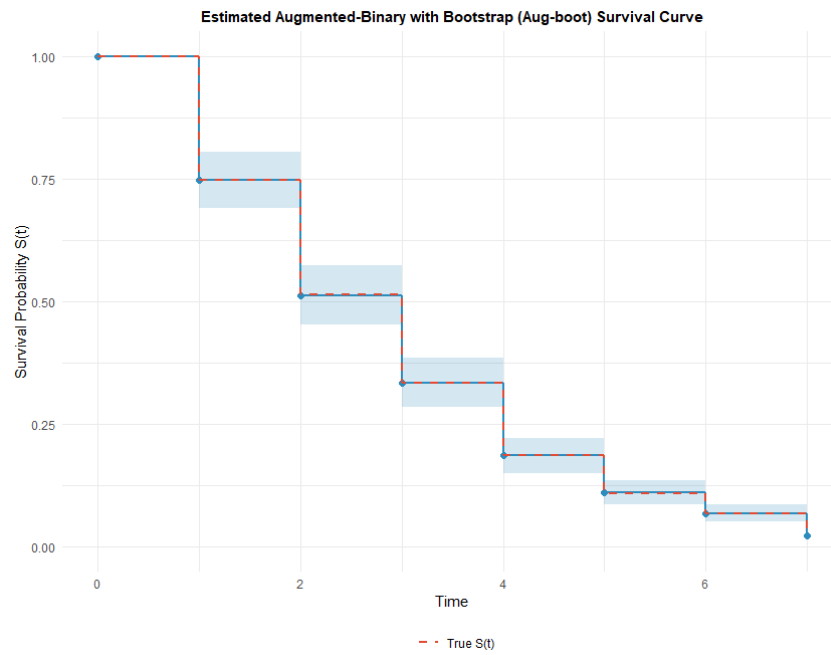


Figure 5.5: Estimated augmented-binary with bootstrap survival curve for 7 time points. Shaded regions are confidence intervals computed from 1000 iterations of 200 bootstrap samples.

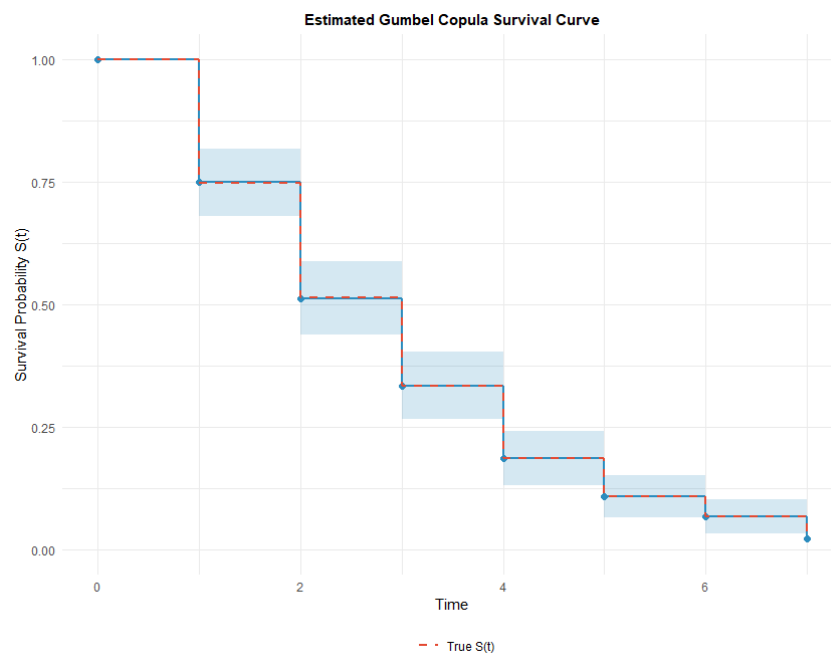


Figure 5.6: Estimated Gumbel copula survival curve for 7 time points. Shaded regions are confidence intervals computed from 1000 iterations of 200 bootstrap samples.

5.3.2 Comparative Trial Results

In the comparative trial, only 5 time points were considered because computation time for 5 time points was much quicker compared to 7 time points. For 5 time points in the single arm trial, the Gumbel copula was the best performing copula so it will be used in this section. To assess the power of the Kaplan-Meier estimate, the log-rank test was used to determine if the Kaplan-Meier estimate could detect a difference between the survival curves of the control and treatment arms. For the augmented-binary with delta method, a pooled Z-statistic was used to determine if this method could detect the difference between the hazard rate of the control and treatment arms. The copula method used the absolute difference in survival curves summed over time points and a permutation to compute the power. All the methods had their power assessed on several scenarios listed in Section 5.2.2. The package `tumourSim` also handles the comparative trial simulations. The results of the power comparisons are listed in Table 5.5 and described below.

The true null scenario tests the Type I error rate, which is the probability of a false positive (incorrectly rejecting the null hypothesis). All the methods for scenario 1 are around the 0.05 target so are well-calibrated to test scenarios 2 to 7. Scenario 2 tests a small treatment effect of $\beta = -0.2$ and $\gamma_{ctrl} = \gamma_{trt} = 0$ which means the tumour size has no effect on the probability of new lesions. In this case, the copula method does offer marginally more power compared with the Kaplan-Meier at 39.5% against 37.1% but the augmented-binary method performs best here with a power of 40.5%. However, when the treatment effect increases in scenarios 3 and 4, the augmented-binary outperforms both the copula and Kaplan-Meier methods with the copula performing worst in those scenarios.

In scenarios 5 to 7, where γ_{ctrl} and γ_{trt} were introduced, the copula marginally outperformed the Kaplan-Meier in all scenarios with power ranging from 67.4% to 99.8% compared with 62.2% to 99.6%. The copula method was also competitive with the augmented-binary method, offering increased power in scenario 6 at 99.3% against 99.0% and was comparable in scenario 7 (99.9% versus 99.8%).

Scenario 8 shows an interesting property of the copula compared to the augmented binary. This scenario uses a heavy tail t-distribution with 4 degrees of freedom to generate the log tumour size ratios so occasional extreme values appear but other parameters are 0 ($\beta = \gamma_{ctrl} = \gamma_{trt} = 0$). We would expect the Type I error rate in this null scenario to be approximately 0.05, which the copula and Kaplan-Meier are, but the augmented-binary method massive over-estimates it. This occurs again when γ_{ctrl} and γ_{trt} are introduced in scenario 10. The augmented-binary method is rejecting the null hypothesis when there is no difference between the arms 17.4% of the time in scenario 8 and 17.0% of the time in scenario 10. This makes a direct comparison with the Kaplan-Meier and copula method difficult in scenario 9 although we can say the Kaplan-Meier outperforms the copula here with a power of 92.4% versus 87.5%.

Table 5.5: Power comparison of log-rank, Aug-delta, and copula methods across simulation scenarios. Results are generated with 200 patients per arm, 1000 iterations and 200 bootstrap samples.

Scenario	Description	Log-rank	Aug-delta	Copula
1	True null (Type I)	0.040	0.045	0.045
2	No gamma, small trt effect	0.371	0.405	0.395
3	No gamma, medium trt effect	0.718	0.795	0.693
4	No gamma, large trt effect	0.912	0.964	0.828
5	Differing gamma, no trt effect	0.622	0.694	0.674
6	Differing gamma, small trt effect	0.974	0.990	0.993
7	Differing gamma, medium trt effect	0.996	0.999	0.998
8	Heavy-tail null + dep. misalignment	0.055	0.174	0.046
9	Heavy-tail H1 + dep. misalignment	0.924	0.864	0.875
10	Heavy-tail null, matched dependence	0.057	0.170	0.044

The power values will have an important role in analysis the results of the statistical comparison of the treatment arms in the colorectal cancer data set in Chapter 6.

Chapter 6

Application to Colorectal Cancer Data

This section will discuss the phase 3 PRIME trial from Amgen to compare treatments for colorectal cancer and describe their process, data collection and trial demographic. Then, the Kaplan-Meier and copula method will be applied to the data, to investigate the difference between the treatment groups and identify if a treatment is effective.

6.1 Data Description and Endpoint Construction

The phase 3 PRIME trial by Amgen compares the treatment of colorectal cancer with the drug FOLFOX4 and FOLFOX4 combined with panitumumab. The dataset is available on <https://data.projectdatasphere.org/projectdatasphere/html/home> with identifier Colorec_Amgen_2006_309 and has previously been analysed by [14] and [31].

The trial contained 1183 patients randomised into two treatment groups. These groups were FOLFOX4 alone and panitumumab with FOLFOX4. However, only 442 patients were considered in the analysis. Due to the number of exclusions, it is worth noting that the results may differ if all participants were included in the analysis.

The two main variables of interest were: (a), the sum of the longest diameters (SLD) which is a continuous longitudinal variable and (b), an indicator as to whether a new lesion was identified at a follow-up appointment. For the copula approach, two progression events were considered: (i), the SLD increasing by more than 20% compared to the minimum of all previous SLDs and (ii), whether a new lesion is present at a follow-up appointment. For the traditional survival analysis with the Kaplan-Meier estimator, we consider a composite progression event which is the first occurrence of either of these events. The 442 patients that were chosen were the subjects that had SLD measurements, new lesion indicators and KRAS type recorded.

As well as treatment group, we will also consider differences in survival curves between KRAS type. KRAS is a gene for making proteins and the PRIME trial considered patients

with either wild-type or mutant KRAS.

Table 6.1 shows the number of patients in each treatment group and gives frequencies of the different progression events.

Table 6.1: Description of colorectal cancer dataset

Treatment	FOLFOX alone	FOLFOX + panit.
Number of patients enrolled	593 (50.1%)	590 (48.9%)
Number of patients for the analysis	223 (50.5%)	219 (49.5%)
Number of Patients with Wild-type KRAS	132 (59.2%)	91 (40.8%)
Number of Patients with Mutant KRAS	128 (58.4%)	91 (41.6%)
Number of Deaths	164 (73.5%)	164 (74.9%)
Number of composite progression events	76 (34.1%)	67 (30.6%)
Number of new lesion events	61 (27.4%)	52 (23.7%)
Number of SLD progression events	20 (9.0%)	18 (8.2%)

Figure 6.1a displays a spaghetti plot of the natural logarithm of SLD over time for each patient stratified by treatment group. The figure includes individual patient trajectories as well as locally estimated regression (loess) curves with $\text{span} = 0.90$ for each treatment group. The smoothed trajectories indicate a general downward trend in $\log(\text{SLD})$ over time in both groups. Figure 6.1b shows a spaghetti plot of the natural logarithm of SLD over time for each patient stratified by KRAS type. We also see a general downwards trend in $\log(\text{SLD})$ for both types of KRAS over the length of the study. The natural logarithm of SLD is used in Figure 6.1 to create a more readable spaghetti plot but the untransformed SLD values will be used in Section 6.2 inline with the RECIST framework.

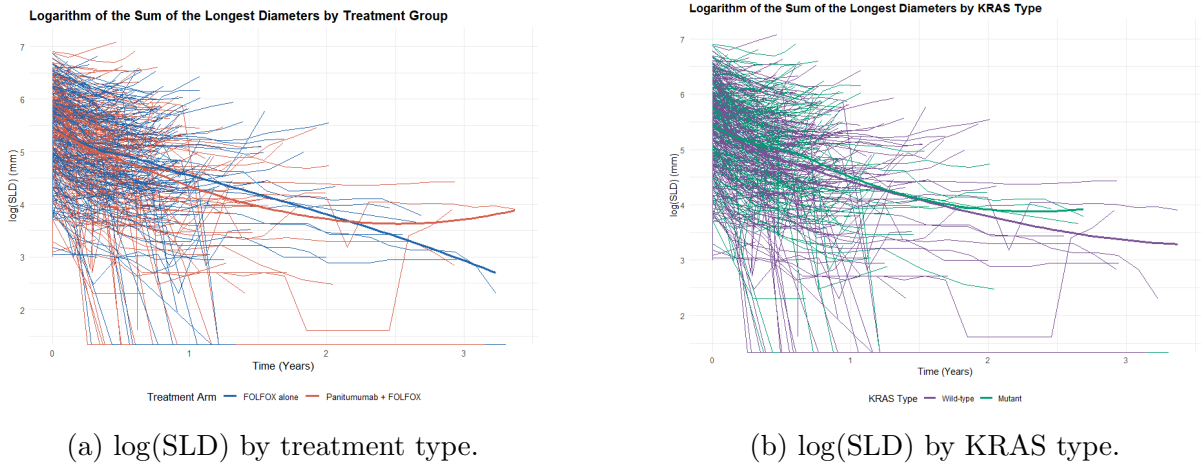


Figure 6.1: Spaghetti plots of the natural logarithm of the sum of the longest diameters by treatment group and KRAS type.

Using the RECIST framework outlined in Section 4.2, applied to the SLD data and new lesion indicators we will obtain the Kaplan-Meier and copula estimates for the progression free survival curves of each treatment group.

6.2 Data Analysis

In order to use the `tumourSim` package and fit a Kaplan-Meier estimator, the progression events defined in Section 6.1 were identified for each patient considered for analysis. We will consider the trial split into 2 groups; one by treatment group and the other by KRAS type to see if either has an effect on survival outcomes.

6.2.1 Treatment-Stratified Analysis

Firstly, the dataset was split into FOLFOX and Panitumumab + FOLFOX treatment groups. A Kaplan-Meier estimates for the new lesion events and the SLD progression events were obtained for each group. These are the marginals of the copula used to estimate the copula-based survival function. Next, three different copulas, Clayton, Frank and Gumbel, were fit to the marginals for each treatment group and using the formula

$$\hat{S}_{PFS}(t) = \hat{S}_{T_D, T_Y}(t, t) = \hat{S}_{KM}^{(D)}(t) + \hat{S}_{KM}^{(Y)}(t) - 1 + C(1 - \hat{S}_{KM}^{(D)}(t), 1 - \hat{S}_{KM}^{(Y)}(t); \hat{\theta}),$$

an estimate for the progression-free survival at time t can be obtained for both groups. Then, the AIC, copula parameter θ and Kendall's τ for each copula are computed.

Table 6.2 shows the values of the AIC, θ , τ and the upper ($\lambda_U^{\tilde{C}}$) and lower tail ($\lambda_L^{\tilde{C}}$) dependence for each type of copula. The AIC was computed by R using the formula described in Section 3.7.2. The value of θ was estimated numerically through maximum likelihood estimation. This involved maximising the log-likelihood of each copula, which was derived in Section 3.8.3. The value of Kendall's τ was also computed by R but uses the relationship between τ and the copula parameter which was derived in Section 3.9.1. Section 3.9.2 shows the relationship between the base copula C (in this case Clayton, Gumbel or Frank) and the survival copula $\tilde{C}$. For example, a base copula with lower tail dependence means the survival copula will have upper tail dependence.

The Clayton copula has the largest AIC for both treatment groups. Also, the upper tail dependence of the survival copula is extremely small which suggests the Clayton copula performs poorly here and should be disregarded. The Gumbel copula has the lowest AIC for the FOLFOX group (AIC=-93.312) and the second lowest AIC for the Panitumumab + FOLFOX (AIC=-46.841) but is competitive with the Frank copula because the difference between the AIC values is small ($\Delta AIC = 2.588$). Also, the most successful copula from the simulation study was the Gumbel. Based on this performance, we will use the Gumbel copula to compare to the Kaplan-Meier estimator. Lower tail dependence in the survival copula corresponds to the lower-left of the joint distribution of survival times, where both endpoints have short times to event. This means the Gumbel copula captures the clinically important pattern in which rapid SLD progression and new lesion appearance tend to occur together in the same patients. The Kendall's τ for the FOLFOX group is 0.256 compared to 0.307 for the Panitumumab + FOLFOX group. Clinically, this implies patients in the FOLFOX + Panitumumab arm show slightly stronger co-occurrence of SLD progression and new lesion events, which could suggest a more aggressive disease trajectory in that group.

Table 6.2: AIC, copula parameter θ , Kendall's τ , and survival-copula tail dependence for the Clayton, Frank and Gumbel copulas fitted to the colorectal cancer dataset, stratified by treatment group.

Quantity	Clayton Copula		Frank Copula		Gumbel Copula	
	FOLFOX	(Panit.+FOLFOX)	FOLFOX	(Panit.+FOLFOX)	FOLFOX	(Panit.+FOLFOX)
AIC	-7.806	-47.846	-49.429	-81.709	-46.841	-93.312
θ	0.058	0.089	2.579	3.225	1.344	1.444
Kendall's τ	0.028	0.043	0.269	0.326	0.256	0.307
λ_L^C	0	0	0	0	0.325	0.384
λ_U^C	6.5×10^{-6}	4.1×10^{-4}	0	0	0	0

For $1 \leq t \leq 7$, M_t represents a 6-month period in the study. This is done for ease of comparison between the copula and Kaplan-Meier estimates because the event time values were in a continuous format and not discrete follow-up times. Table 6.3 and Table 6.4 display the estimated mean survival rates, 95% confidence interval widths for the FOLFOX and Panitumumab + FOLFOX treatment groups respectively.

The Gumbel copula consistently estimates a slightly higher mean survival rate across all time periods in both treatment groups compared to the Kaplan-Meier. During the first time period M_1 , the Gumbel copula offers the biggest reduction in confidence interval width in both treatment arms with a reduction of 21.78% and 27.78%. Further reductions are also noted in the time periods M_4 to M_7 in the FOLFOX group and at M_4 in the Panitumumab + FOLFOX group.

Table 6.3: Estimated mean survival times, CI widths, and changes in CI width for the Kaplan-Meier and Gumbel copula for the FOLFOX alone treatment group across periods M_t . The 95% confidence intervals for the copula method are obtained by 200 bootstrap samples.

Time Period	Estimated Mean $S(t)$		CI Width		Change in CI Width
	KM	Copula	KM	Copula	% Change
M_1	0.807	0.870	0.101	0.079	-21.78
M_2	0.464	0.490	0.147	0.160	8.84
M_3	0.165	0.181	0.134	0.149	11.19
M_4	0.107	0.114	0.122	0.112	-8.20
M_5	0.097	0.106	0.123	0.106	-13.82
M_6	0.097	0.106	0.123	0.106	-13.82
M_7	0.097	0.106	0.123	0.106	-13.82

Table 6.4: Estimated mean survival times, confidence intervals, and CI widths for the Kaplan-Meier and Gumbel copula for the Panit.+FOLFOX treatment group across periods M_t . Missing values (“–”) occur when the estimated mean survival probability is 0.000.

Time Period	Estimated Mean $S(t)$		CI Width		Change in CI Width
	KM	Copula	KM	Copula	% Change
M_1	0.789	0.872	0.108	0.078	-27.78
M_2	0.428	0.474	0.158	0.165	4.43
M_3	0.194	0.269	0.159	0.162	1.89
M_4	0.108	0.177	0.169	0.164	-2.96
M_5	0.067	0.135	0.184	0.189	2.72
M_6	0.000	0.088	–	0.185	–
M_7	–	0.080	–	0.172	–

Figure 6.2 displays the Kaplan-Meier and Gumbel copula survival curves for the colorectal cancer dataset. These graphs use the actual event times instead of the grouped time periods like in Table 6.3 and Table 6.4 to get a full view of the survival curves. Both plots follow a similar structure, with treatment groups having close overlapping curves at the start of the study which decrease quickly until 1 year has passed. Then, both curves level off and their gradient decreases and this trend continues until the end of the study. We do see a difference between the Kaplan-Meier curves and the Gumbel copula curves. The Kaplan-Meier curves remain close together throughout the trial whereas the Gumbel curve sees a larger separation between treatment groups from year 1 to year 2. This suggests that the copula method is identifying an underlying structure that the Kaplan-Meier estimate does not.

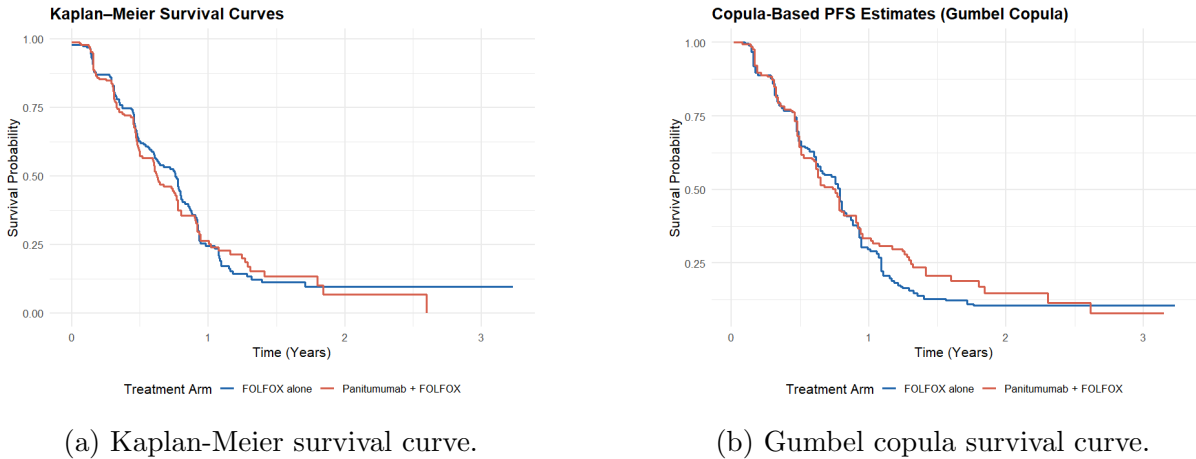


Figure 6.2: Comparison of the Kaplan-Meier and Gumbel copula survival curves for the colorectal cancer dataset, stratified by treatment group.

To formally assess whether the survival curves differ between treatment arms, we apply the log-rank test on the Kaplan-Meier curves and a permutation test based on the integrated absolute difference (L_1 statistic) on the copula curves.

Table 6.5 shows the test statistics and corresponding p-values for each method. The null hypothesis is that the estimated survival curves of the two treatment arms are equal ($\hat{S}_{\text{FOLFOX}} = \hat{S}_{\text{Panit.}+\text{FOLFOX}}$). The Kaplan-Meier method has a p-value of $0.666 > 0.05$ and the Gumbel copula approach has a p-value of $0.262 > 0.05$. Neither of these tests gives a p-value less than 0.05 so both methods were unable to find a statistically significant difference between treatment groups at the 5% significance level and we accept the null hypothesis.

Table 6.5: Test statistics and p-values for comparing survival curves. The Kaplan-Meier and the Gumbel copula p-value are generated by 2000 permutations bootstrap samples.

Method	Test Statistic	p-value
Kaplan-Meier (Log-rank)	0.186	0.666
Gumbel Copula (L_1 Permutation)	0.133	0.343

This is consistent with Figure 6.2 because the survival curves in both plots do not experience a clear or sustained separation throughout the trial. The Gumbel copula pp p-value (0.343) is just under half that of the log-rank test (0.666). Although neither reaches significance here, the consistent direction matches simulation findings (Table 5.5, scenarios 2 and 6) where the copula method showed higher power than log-rank under small treatment effects, with or without dependence between tumour size and new lesions.

6.2.2 KRAS-Stratified Analysis

In this analysis, the trial will be split up into the two KRAS types, wild-type and mutant, then we can analyse the effect of each KRAS-type on each treatment method.

Similarly to Table 6.2, the AIC, copula parameter, Kendall's τ and the tail dependence of each copula are presented in Table 6.6. The table is split up into 4 sections based on KRAS-type and treatment group to give the most informed analysis into how KRAS effects treatment.

Table 6.6: AIC, copula parameter θ , and Kendall's τ for the Clayton, Frank, and Gumbel copulas fitted to each KRAS $\times$ treatment cell. The lowest AIC in each cell is highlighted in bold.

Cell	Family	θ	Kendall's τ	AIC
Wild-type / FOLFOX alone	Clayton	0.085	0.041	-11.684
	Gumbel	1.500	0.333	-46.995
	Frank	3.763	0.369	-52.147
Wild-type / Panitumumab + FOLFOX	Clayton	0.068	0.033	-8.756
	Gumbel	1.407	0.289	-37.591
	Frank	3.219	0.326	-41.500
Mutant / FOLFOX alone	Clayton	0.040	0.020	-0.472
	Gumbel	1.302	0.232	-14.141
	Frank	2.426	0.255	-15.738
Mutant / Panitumumab + FOLFOX	Clayton	0.086	0.041	-26.339
	Gumbel	1.483	0.326	-54.151
	Frank	4.093	0.395	-57.257

The Frank copula achieves the lowest AIC in each of the four cells. We retain the Gumbel copula for the remaining analysis for two reasons. First, the AIC differences are small in absolute terms, particularly relative to the magnitude of AIC values themselves and the Gumbel copula is consistently competitive in every cell. Second, Gumbel copula was identified in the simulation study (Section 5.3) as the most appropriate copula for RECIST-style endpoints based on its lower tail dependence, which captures the clinical pattern of co-occurring early progression events and aligns with the bivariate dependence structure observed in the treatment-stratified analysis (Section 6.2.1). Using a single copula family across all subgroup analyses ensures that observed differences in θ and τ between cells reflect genuine differences in dependence rather than artefacts of family selection.

Figure 6.3 shows the Kaplan-Meier survival curves for the wild-type and mutant KRAS with the curves being colour-coded by treatment group. We can clearly identify a large separation between the curves in the mutant KRAS plot between 1 and 2 years that is not present in the wild-type plot. The wild-type plot consistently overlaps across the duration of the study and shows no clear separation at any point.

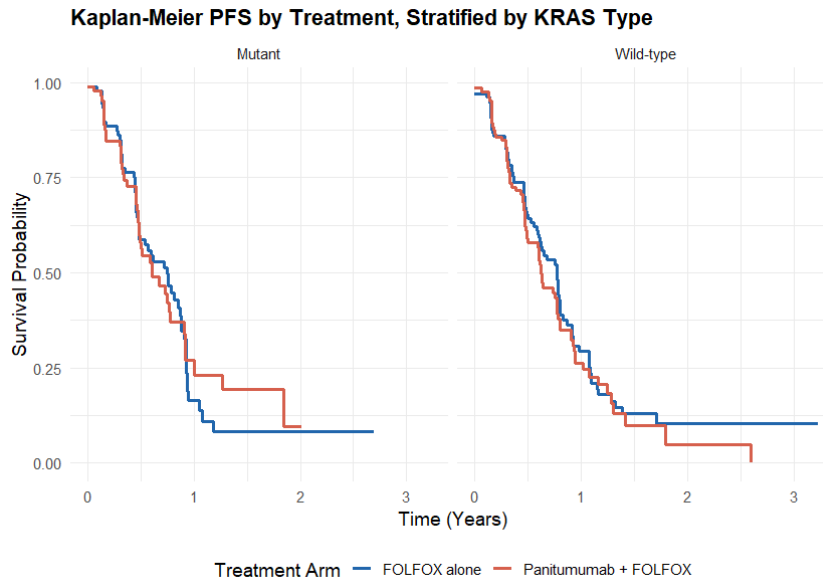


Figure 6.3: Kaplan-Meier estimate of

Figure 6.4 shows the Gumbel copula survival curves for the wild-type and mutant KRAS. A similar trend to Figure 6.3 can also be identified with a big separation between treatment groups in the mutant KRAS plot.

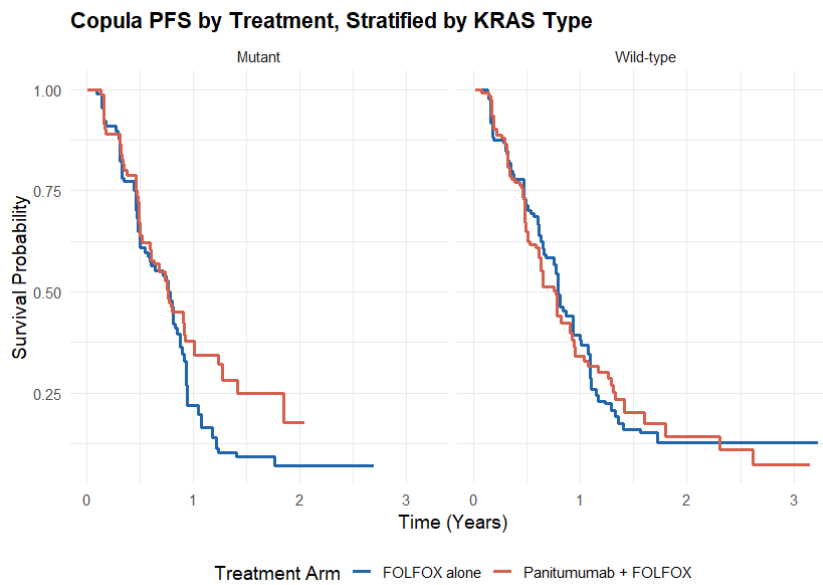


Figure 6.4: Caption

Again, to formally assess whether the survival curves differ between treatment arms within each KRAS-type, we apply the log-rank test on the Kaplan-Meier curves and a permutation test based on the integrated absolute difference (L_1 statistic) on the copula curves.

Table 6.7 shows the test statistics and p-values for comparing the FOLFOX alone and

Panitumumab + FOLFOX survival curves within each KRAS stratum. In the wild-type subgroup, neither method detected a significant difference at the 5% level ($p = 0.455$ for the log-rank, $p = 0.761$ for the copula permutation), consistent with the closely overlapping survival curves observed in Figures 6.3 and 6.4. In the mutant subgroup, the two methods reach different conclusions: the log-rank test finds no evidence of a treatment effect ($\chi^2 = 0.023$, $p=0.879$), whereas the copula L_1 permutation test detects a highly significant difference ($L_1 = 0.411$, $p = 0.008$).

Table 6.7: Statistical comparison of treatment arms (FOLFOX alone versus Panitumumab + FOLFOX) within each KRAS stratum. The Kaplan-Meier curves are compared using the log-rank test; the Gumbel copula curves are compared using a permutation test based on the integrated absolute difference (L_1 statistic), with 2000 permutations.

KRAS Stratum	Method	Test Statistic	p -value
Wild-type	Kaplan-Meier (log-rank)	0.559	0.455
	Gumbel copula (L_1 permutation)	0.112	0.761
Mutant	Kaplan-Meier (log-rank)	0.023	0.879
	Gumbel copula (L_1 permutation)	0.411	0.008

The discrepancy is not accidental. Looking at the survival curves shows that the two arms overlap closely until approximately one year, after which Panitumumab + FOLFOX maintains a sustained percentage-point survival advantage through year 2 of approximately 10%. The event counts are consistent with this trend with 45 of 91 Mutant patients experienced an event under Panitumumab + FOLFOX (49.5%) compared with 58 of 91 under FOLFOX alone (63.7%), a 14 percentage-point difference in event rate over the same period of time. The implications of this will be reviewed in the discussion.

6.3 Discussion

When the data set was stratified by treatment group (FOLFOX alone and Panitumumab + FOLFOX), neither the Kaplan-Meier nor Gumbel copula method (which was chosen based on the lowest AIC values) detected a statistically significant difference between the survival curves at the 5% significance level ($p = 0.666$ and 0.343 respectively). Despite neither test producing a statistically significant p -value, the copula p -value was just under half of the log-rank p -value which matches the scenarios in Table 5.5 where the copula had a slightly higher power than the Kaplan-Meier for small treatment effects (scenario 2 and 5) with or without tumour-lesion dependence. Furthermore, the copula method provides additional information, inaccessible to the Kaplan-Meier, about the dependence between the tumour sizes and new lesions through the Kendall's τ and tail dependence. The Gumbel copula identified $\tau = 0.256$ in the FOLFOX alone arm and $\tau = 0.307$ in the Panitumumab + FOLFOX arm. Through the relationship between the Gumbel copula and its associated survival copula, this corresponds to lower-tail dependence on the survival-time scale and both values indicate moderate positive dependence, meaning patients with early tumour progression tend to also experience early lesion appearance.

This is consistent with shared underlying disease biology.

The data was then divided into different groups based on KRAS-type and then the Kaplan-Meier and copula survival curves were estimated for each treatment in each group. There was no difference detected by either method in the wild-type subgroup ($p = 0.455$ for the log-rank, $p = 0.761$ for the copula permutation). This result differed from the published PRIME trial findings which found that Panitumumab + FOLFOX had a significant progression-free survival benefit compared to FOLFOX alone for the KRAS wild-type. The published PRIME trial findings reported a significant progression-free survival benefit for Panitumumab + FOLFOX over FOLFOX alone in the wild-type subgroup (median PFS 9.6 vs 8.0 months, $HR = 0.80$, $p = 0.02$) [12]. This discrepancy is likely due to the smaller subset of patients ($n = 442$) out of the full 1100 patients that were used in the full PRIME trial analysis and the different definition of progression which did not take into account death. In the mutant subgroup, there was another result that differed from the published PRIME findings. The copula permutation test suggested evidence of a difference between treatment groups in the mutant subgroup with $p=0.008$ but the log-rank test did not detect a difference with $p=0.879$. Importantly, this does not imply evidence against existing clinical guidance, but rather demonstrates that the copula-based framework may detect survival-curve differences that are less apparent under standard log-rank weighting schemes. Two complementary explanations help interpret this result.

The discrepancy between the two tests reflects how each handles non-proportional hazards. The log-rank test weights event times by the at-risk set size, which concentrates statistical power early where the curves in Figure 6.3 and Figure 6.4 overlap. The late divergence between years 1 and 2 contributes little because few patients remain at risk by then. The L_1 statistic weights time points equally on the survival probability scale, capturing the late difference that log-rank misses. This is a well-known limitation of log-rank tests under non-proportional hazards [18] and shows the value of permutation tests.

However, the clinical interpretation requires more explanation. The published PRIME trial results reported no Panitumumab benefit in the Mutant subgroup, which forms the basis of current treatment guidelines recommending Panitumumab only for KRAS Wild-type patients. Several factors may contribute to the discrepancy between our finding and the published result. Firstly, the public dataset comprises only 442 of the original KRAS-typed PRIME patients which was approximately 1100, for whom longitudinal tumour and lesion data were available. Secondly, the endpoint here is restricted to RECIST tumour progression and new lesion appearance, excluding death and other forms of clinical progression captured by PRIME’s investigator-assessed PFS endpoint. Thirdly, the sample size in the mutant subgroup ($n = 91$ per arm) is small relative to the published trial. The fact that the L_1 permutation test detects late divergence which the log-rank misses is robust to these caveats, as it depends only on the shape of the survival curves observed in this dataset rather than on whether those curves represent the underlying clinical truth.

Chapter 7

Conclusions

In this dissertation, we presented a copula-based survival function to estimate the survival function of events defined by: (i) the progression of a continuous variable past a threshold and, (ii) a binary indicator, as an alternative to the Kaplan-Meier estimator. In particular, the model handles the RECIST progression framework which appears in clinical trials relating to solid tumour cancers such as colorectal cancer. Additionally, an R package was developed to handle estimating the marginal distributions of the copula, the copula parameter θ and the copula-based survival function. The Gumbel copula was identified in the simulation study as the most appropriate copula for this setting.

In the single arm simulation trial, we showed the Gumbel copula with upper tail dependence was the most appropriate copula across 5 and 7 time points, even after changing the mean vector and covariance matrix. This suggests that bivariate progression-free survival endpoints under the RECIST framework may generally exhibit lower tail dependence on the original time scale (via the relationship between the base copula and survival copula derived in Section 3.9.2), a hypothesis that would benefit from validation across other RECIST-evaluated trials. Lower tail dependence on the survival-time scale implies a co-occurrence of early event times: patients who experience early tumour progression also tend to experience the appearance of new lesions early on, and vice versa. This is consistent with a clinical setting where aggressive disease drives both tumour growth and new lesion formation through a shared biological mechanism. This pattern justifies our choice of the Gumbel copula over the Clayton and Frank copulas, which do not share this tail dependence structure.

When compared with the Kaplan-Meier and augmented-binary methods across 5 and 7 time points in the single arm simulation study, the proposed method offered a high coverage rate around the nominal 95% target and narrower confidence intervals than the Kaplan-Meier. However, the augmented-binary method did have the narrowest confidence intervals and a comparable coverage rate. The Kaplan-Meier is fully non-parametric and only uses the event and censoring information for the endpoint of interest. This makes it robust but inefficient. It does not consider the joint structure between tumour and lesion progression, as well as the underlying continuous tumour-size measurements which leads to wider confidence intervals. The copula-based approach uses a two-step approach which fits the marginals with a Kaplan-Meier estimate and then estimates the copula parameter

θ , so the uncertainty in the marginals from the non-parametric estimator leads to more uncertainty in the final estimated survival rate. The augmented-binary does not have this issue as it uses the continuous longitudinal measurements instead of the progression status which is defined by the dichotomisation threshold. Both the copula and augmented-binary method use the bootstrap method to construct their confidence intervals from the empirical bootstrap distribution, so they do not rely on the normal approximation assumption that the Kaplan-Meier confidence intervals use which is another reason for narrower confidence intervals in these methods. The augmented-binary method does give more precise confidence intervals compared to the joint survival model, but the copula does offer extra insight into the tail dependence structure of the data and a useful clinical interpretation how the two endpoints behave jointly under aggressive disease. We then extended the comparison of the methods in a power analysis.

In the comparative simulation trial, we saw varied success of the copula method compared to the Kaplan-Meier and augmented-binary estimates. All three methods controlled Type I error in the null test (Scenario 0: Kaplan-Meier 0.040, augmented-binary 0.045, copula 0.045). The proposed outperformed the Kaplan-Meier's power when there was a small treatment effect, regardless of whether the lesion model included a coefficient for tumour size but did not reach the augmented-binary method's level. All methods saw an increase in power as the treatment effect grew, with or without an interaction term γ . Under scenarios with no dependence between the tumour and lesion progression endpoints (Scenarios 1–3), the augmented-binary method achieved the highest power, followed by the Kaplan-Meier and then the copula method.

This occurs because when $\gamma = 0$, there is no dependence between new lesions and tumour growth, and the Kaplan-Meier marginals of the copula method are independent and the copula parameter θ offers no extra information to increase the power. The copula method becomes similar to the regular Kaplan-Meier estimate just with extra uncertainty from the two-step estimation procedure which is why it underperforms in these scenarios. When a non-zero γ is introduced (Scenarios 4–6), the copula method's power increased substantially and exceeded that of the Kaplan-Meier, narrowing the gap with the augmented-binary method. In these scenarios, the copula's marginals become dependent on each other and the logistic regression model is a function of previous tumour size. The copula parameter θ captures this dependence and feeds back into the model which improves the estimate and power compared to the Kaplan-Meier. Up to this point the comparison has assumed correct specification of the augmented-binary method's parametric assumptions. We now examine what happens when those assumptions are violated.

In scenarios 8 and 10, we have a null case to test for Type I error but a heavy tail t distribution with $\nu = 4$ which generates the log tumour size ratios. The augmented-binary method exhibits inflated Type I error (0.174 and 0.170 against the nominal 0.05). The augmented-binary method assumes the log tumour size ratios have a normal distribution so a model misspecification violates this assumption. The augmented-binary method's likelihood-based test statistic relies on Gaussian assumptions for both the point estimator and its standard error. Under heavy-tailed dynamics, the model-based variance estimator underestimates the true sampling variability, inflating the test statistic and leads to over-rejection of the null hypothesis. The Kaplan-Meier and copula approaches use the time-to-event data and not the actual longitudinal measurements and so is unaffected by the distribution of the log tumour size ratios. They have Type I error around the nominal

0.05 target (scenario 8: 0.055 and 0.046, scenario 10: 0.057 and 0.044). This means in scenario 9 where we also have a heavy tail t distribution, the power of the augmented-binary method may become unstable or unreliable. The Kaplan-Meier and copula both performed better here, highlighting a key limitation of the augmented-binary method.

Gaussian and Student-t distributions were not considered for modelling as they impose restrictive assumptions on marginal distributions and dependence structure. In particular, Gaussian copulas assume symmetric dependence fully characterised by linear correlation, while Student-t copulas allow heavier tails, they still have symmetric tail dependence. These assumptions are often unrealistic in clinical trial data where measurements may be skewed, bounded, or discrete, and where progression events may exhibit asymmetric and tail-dependent behaviour. Furthermore, initial testing of the Gaussian copula and t-copula resulted in poor coverage rates and lengthy computational time for few iterations. This was due to multiple parameters that needed estimating such as the correlation or degrees of freedom for the t-copula. Therefore, they were not considered for further analysis.

When we have a well-specified Gaussian distribution (scenarios 1 to 7) and are confident in the assumptions due to prior observations or experiments, the augmented-binary method is the best choice. However, when we have a non-Gaussian distribution or are unsure of the tumour dynamics, it may be appropriate to choose the copula model as the non-parametric marginals, parametric dependence and bootstrap variance estimation, at the cost of a modest reduction in power, give an accurate estimator of the survival function for bivariate endpoints.

The PRIME trial from which the colorectal cancer data set was collected involved 1183 patients but only 521 had new lesion indicators available and then, only 442 had these and KRAS type recorded, so any comparison is underpowered when compared to the published trial results. This subset of the data was taken because the copula method required the sum of the longest diameters or tumours and new lesion indicators in a longitudinal format. The endpoints were also re-engineered and constructed from raw measurements and indicators, not including death as a progression event, which differs from the investigator-assessed definition of progression-free survival in the PRIME trial. The results of the statistical tests to identify a difference in survival curves gave some interesting findings which were discussed in depth in Section 6.3. The surprising results may be explainable by the following limitations.

The presented analysis is subject to several limitations. Firstly, a bivariate copula is used and restricted to two endpoints, tumour and lesion progression, where real clinical trials often involve several variables such as death or other biomarkers. Extending the copula to a multivariate setting would be beneficial, at the cost of additional assumptions and much larger samples sizes. Secondly, we have only considered three Archimedean copulas with upper tail dependence (Gumbel), lower tail dependence (Clayton) and no tail dependence (Frank) and one copula parameter θ . This covers a useful range of dependence structures but does not consider every dependence structure. Applying different copulas to this RECIST framework may also be beneficial. Thirdly, the bootstrap method for constructing confidence intervals and the permutation approach for hypothesis testing is computationally expensive, especially for a large number of bootstrap samples and complex copula maximum likelihood estimation. This may limit the practicality of the

copula method when extending it to larger data sets. Fourthly, the colorectal cancer data set uses a non-random subset of the original PRIME trial, where patients were chosen based on availability of measurements so may not be fully representative of the full trial population and the analysis is underpowered for detecting realistic treatment effects. Finally, the simulation study relies on a Gaussian distribution for the log tumour size ratios (scenarios 1 to 7) or a heavy tailed t distribution (scenarios 8 to 10). This covers a wide range of situations and is a reasonable assumption to make but real cancer trials may exhibit other distributions and the performance of the methods may change based on this.

There are a few ways this work could be extending in the future. The first extension worth following up on is whether other RECIST-style frameworks in the bivariate setting exhibit lower tail dependence. This would require analysis of other cancer data sets that follow RECIST. Secondly, extending to three or more marginals in the copula could added extra insight into survival rate prediction, although larger data sets would be required. Finally, further investigation into the Kendall's τ and the coefficients of tail dependence in the colorectal cancer data could explore if they have prognostic value into subgroup identification or stratification that KRAS-type alone provides.

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